Intro to spatial omics

04/22/2025

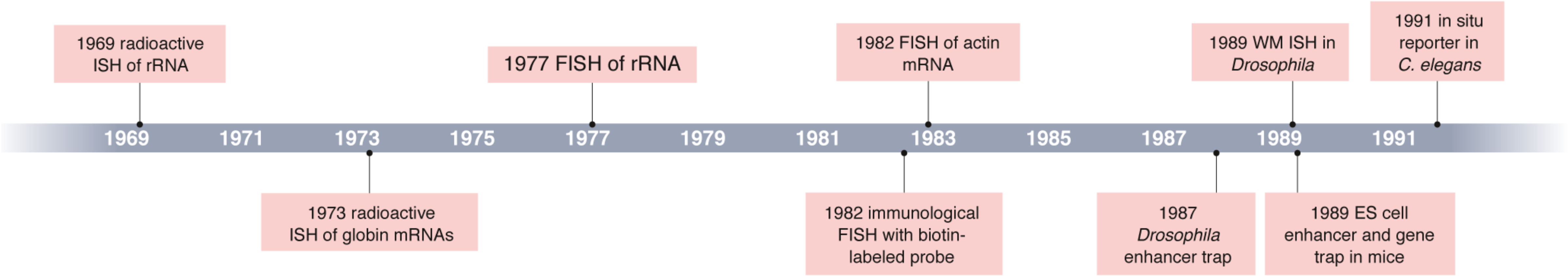
Jiwoon Park, Ph.D.

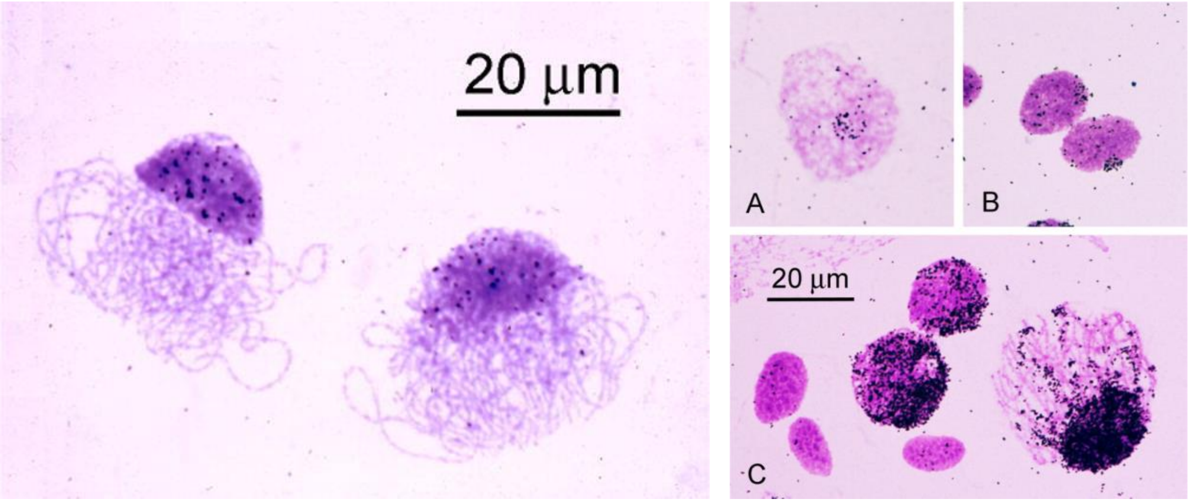
We want the whole platter and their interactions

Bulk	Single cell	Spatial
		
Counts/group	Counts/cell	Counts/cell + location

Before spatial “omics”

Major events in evolution of prequel techniques

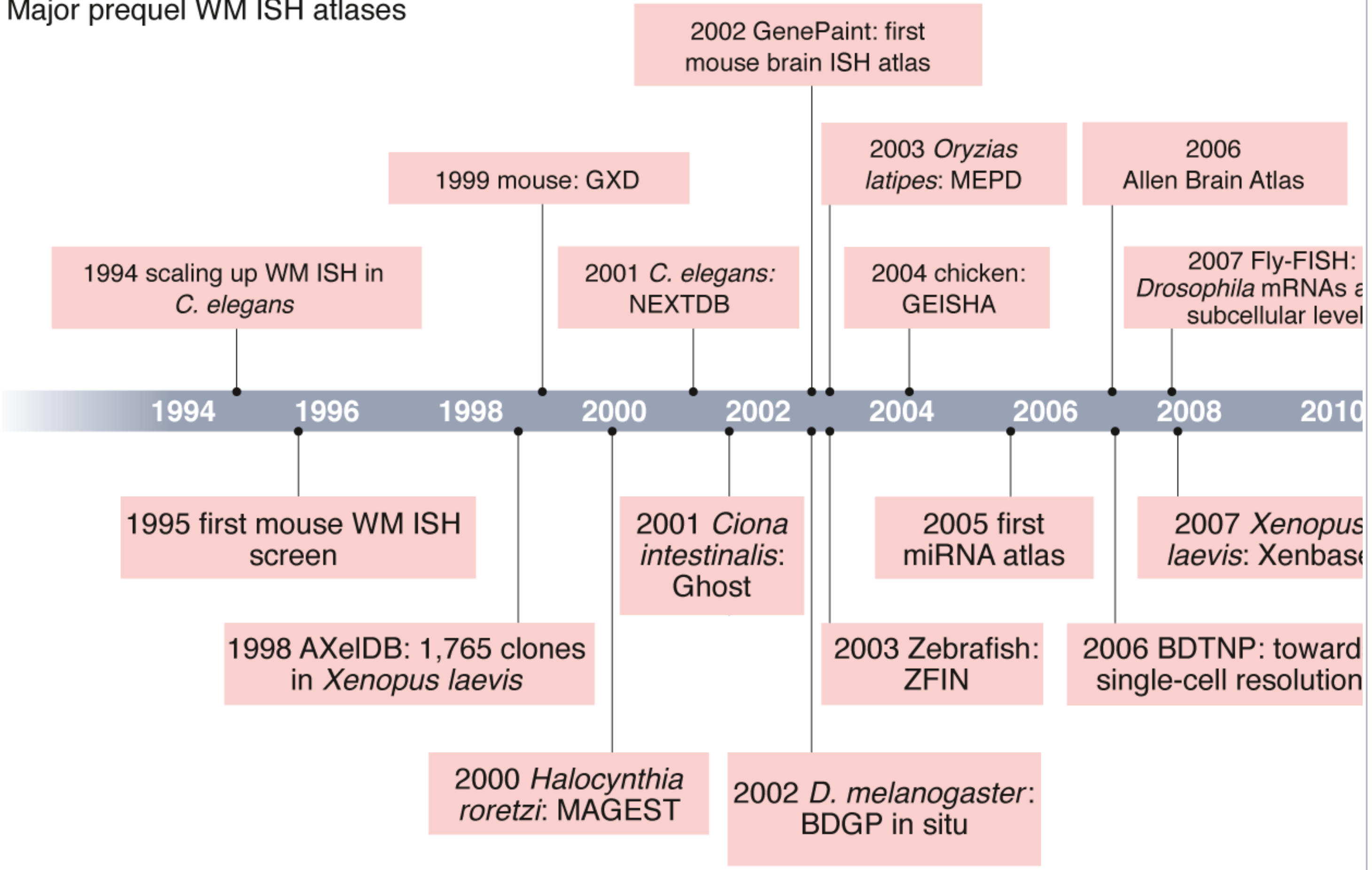




Moses, L., Pachter, L. *Nat Methods* 2022; Image from Gall JG. *Methods*. 2016

Molecular biology with in-situ hybridization

Major prequel WM ISH atlases

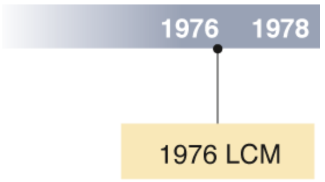


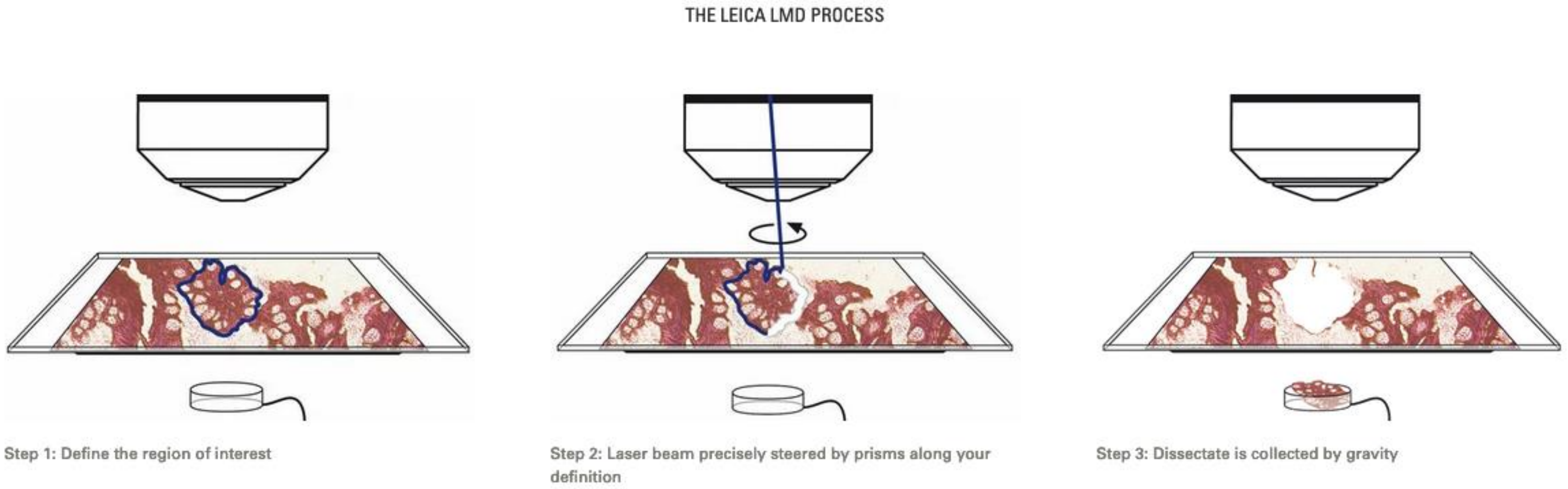
Moses, L., Pachter, L. *Nat Methods* 2022; image from Allen Brain Atlas (https://mouse.brain-map.org/experiment/show/71250306)

Sequencing and multiplexing

Major events in evolution of current-era techniques

2013 high-



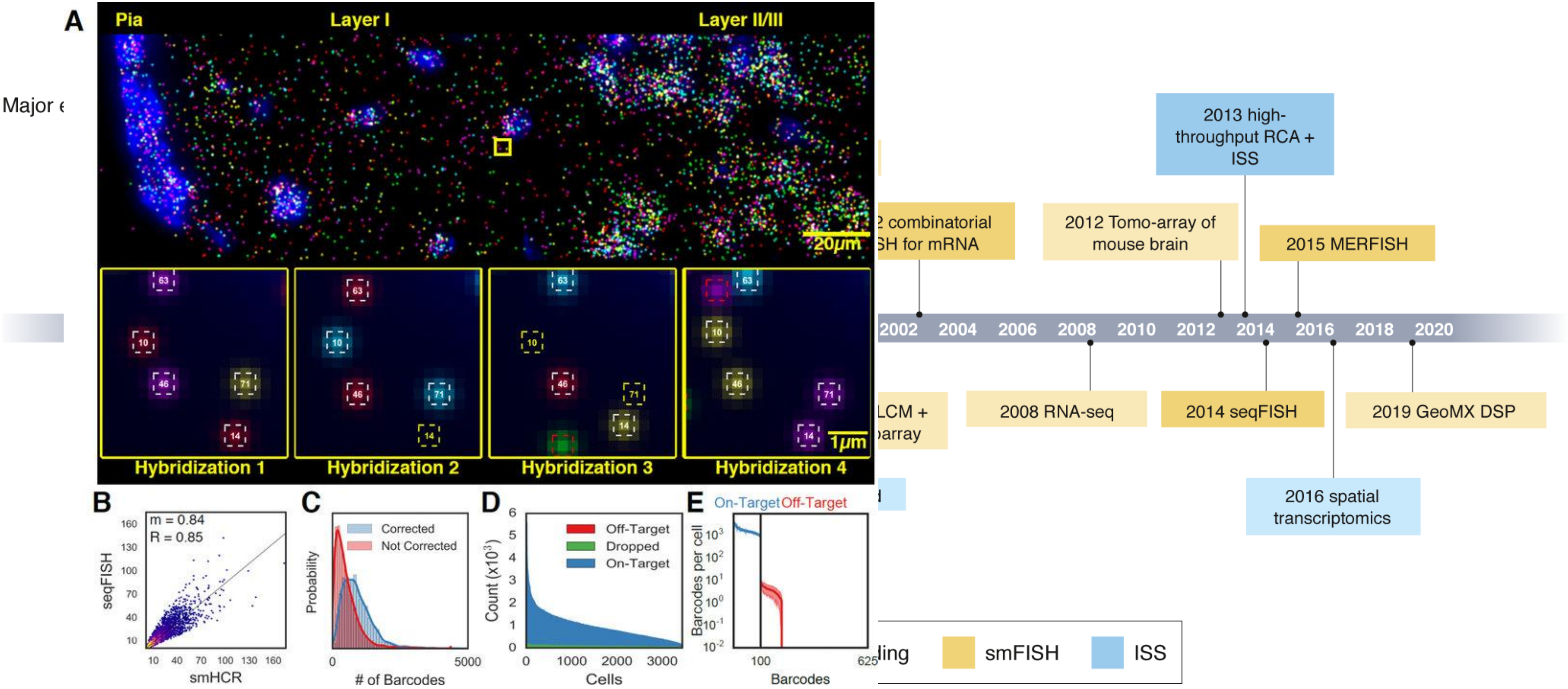


barcoding

Prequel	ROI selection	NGS barcoding	smFISH	ISS

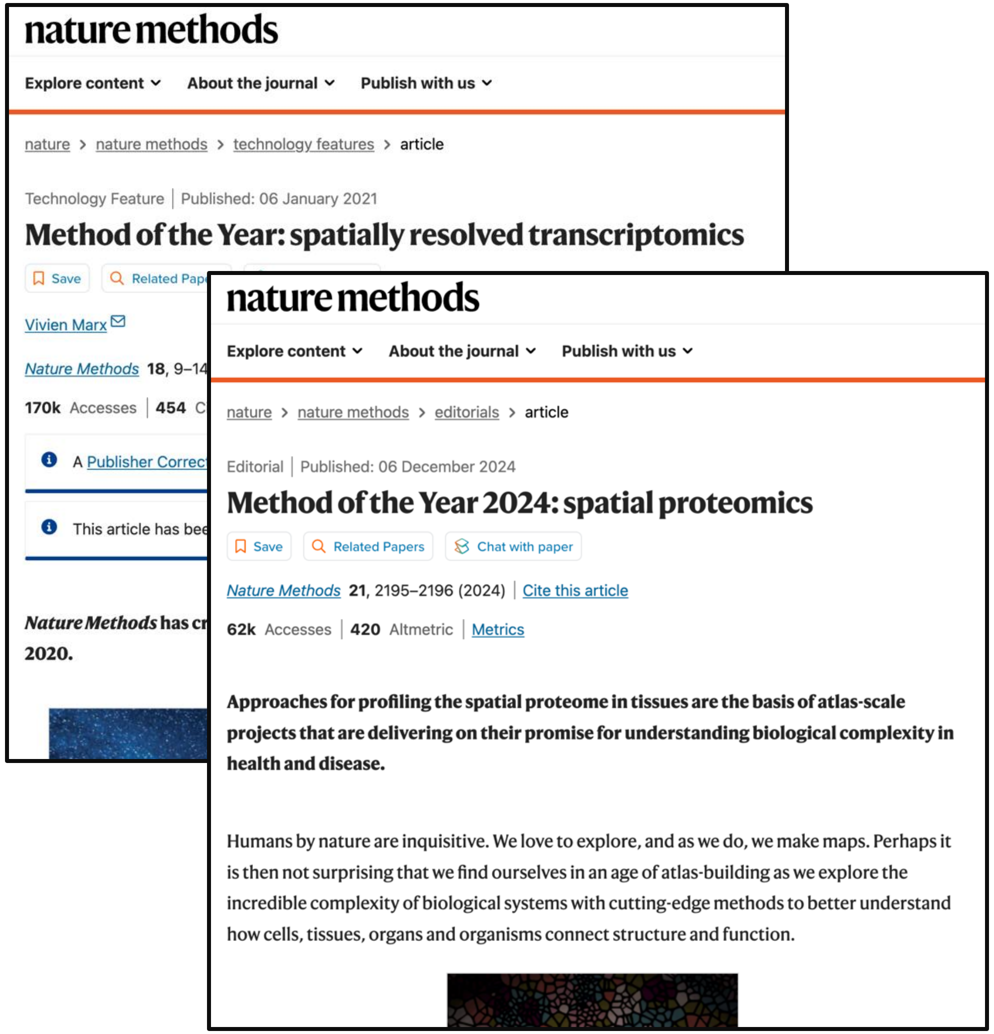
Moses, L., Pachter, L. *Nat Methods* 2022; image from Leica

Sequencing and multiplexing



Moses, L., Pachter, L. *Nat Methods* 2022; image from Shah, Sheel et al. *Neuron* 2016.

Explosion since 2020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Editorial | Published: 06 December 2024

Method of the Year 2024: spatial proteomics

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Nature Methods 21, 2195–2196 (2024) | Cite this article

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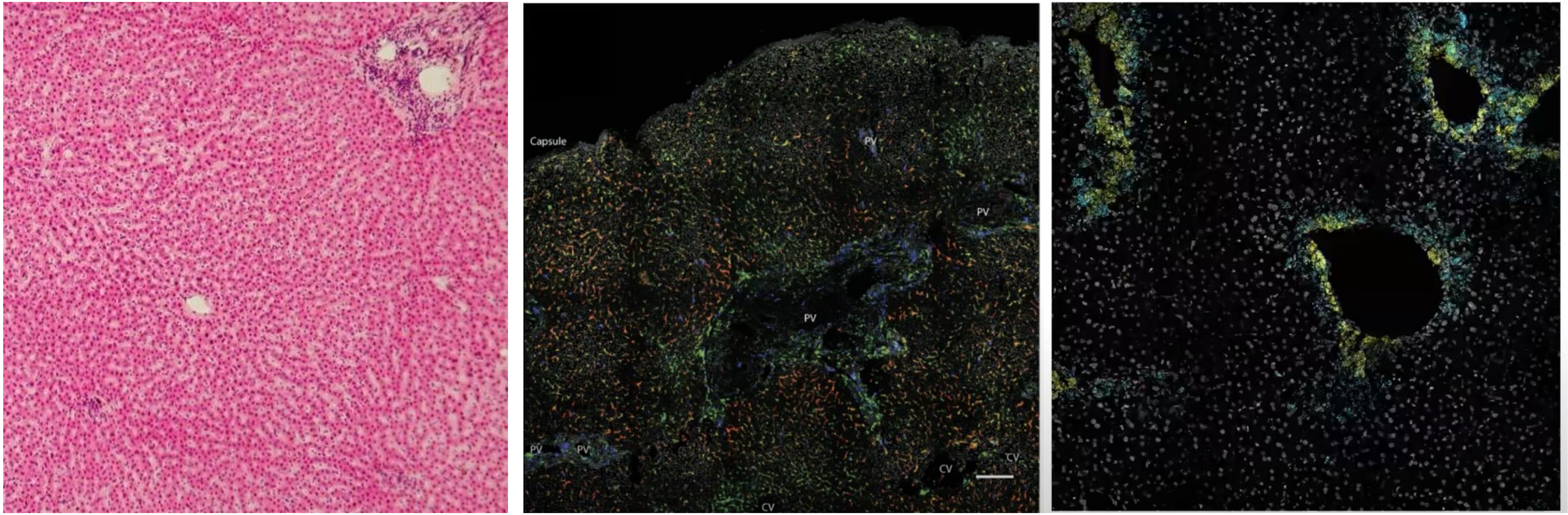
Approaches for profiling the spatial proteome in tissues are the basis of atlas-scale projects that are delivering on their promise for understanding biological complexity in health and disease.

Humans by nature are inquisitive. We love to explore, and as we do, we make maps. Perhaps it is then not surprising that we find ourselves in an age of atlas-building as we explore the incredible complexity of biological systems with cutting-edge methods to better understand how cells, tissues, organs and organisms connect structure and function.

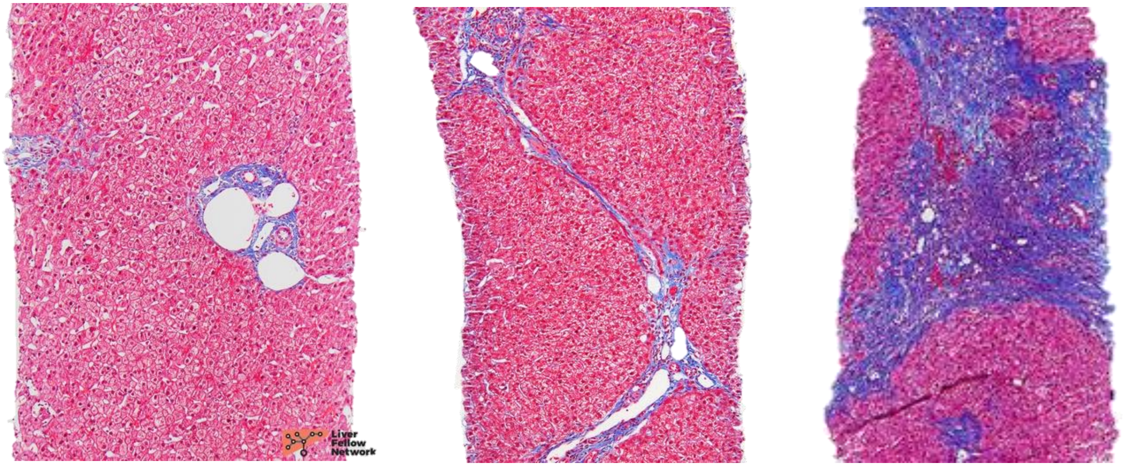
Image from LatchBio

Every organ has a spatial organization

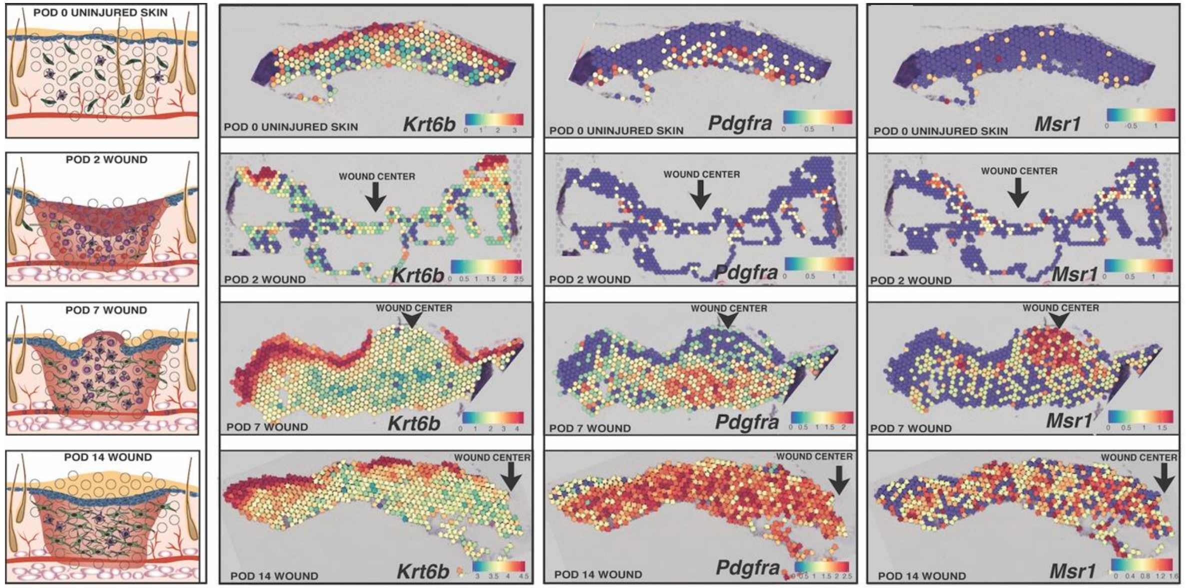


Liver; Resolve Molecular Cartography Images from M Guilliams et al., *Cell* 2022.

Spatial organization of cell types and expressions matter



Liver fibrosis development

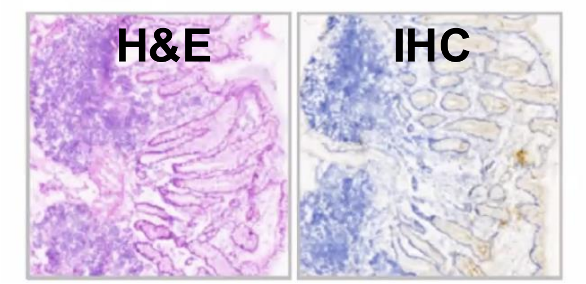


Wound healing and scar formation

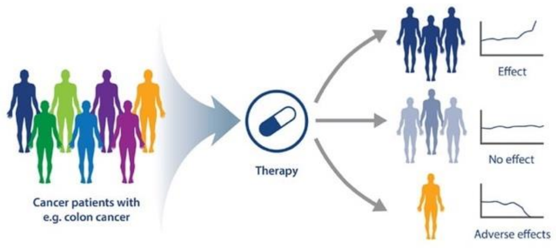
Liver image from AASLD liver fellow network; Foster DS *et al.*, *PNAS* 2021

Spatial omics created opportunities for digital pathology

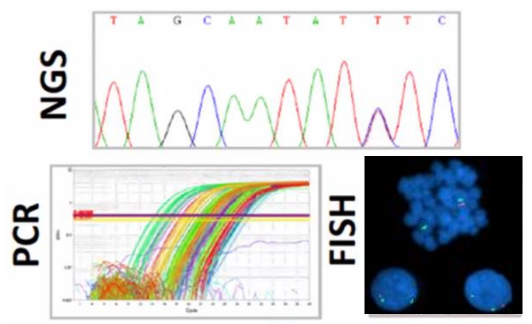
19th Century **Histopathology**



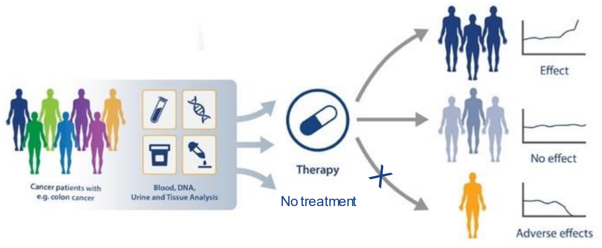
Tissue morphology,
TNM staging/grading



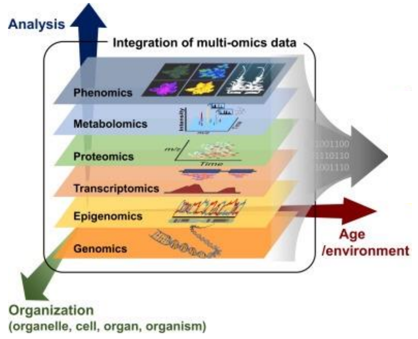
21st Century **Molecular Pathology**



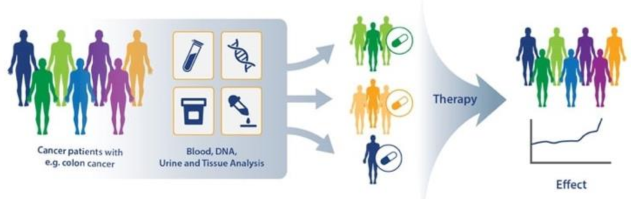
Companion diagnosis



NOW & FUTURE **Digital Pathology**

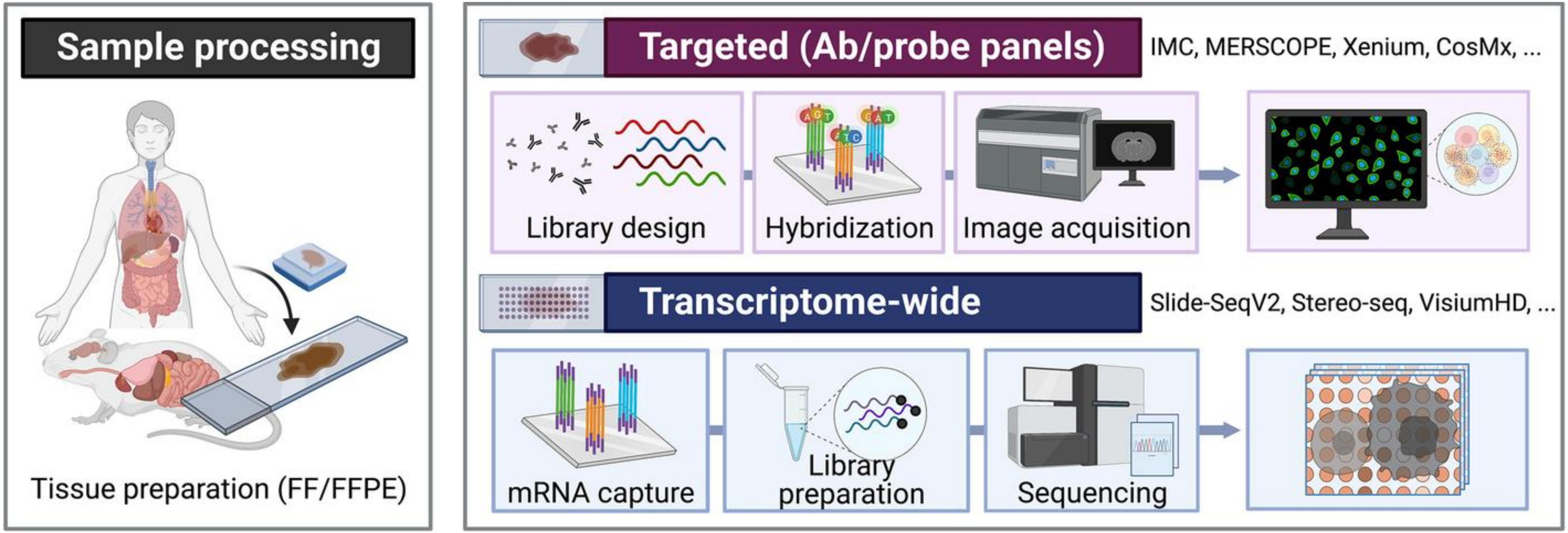


Spatial + Multi omics



One treatment for all → *Personalized diagnostics*

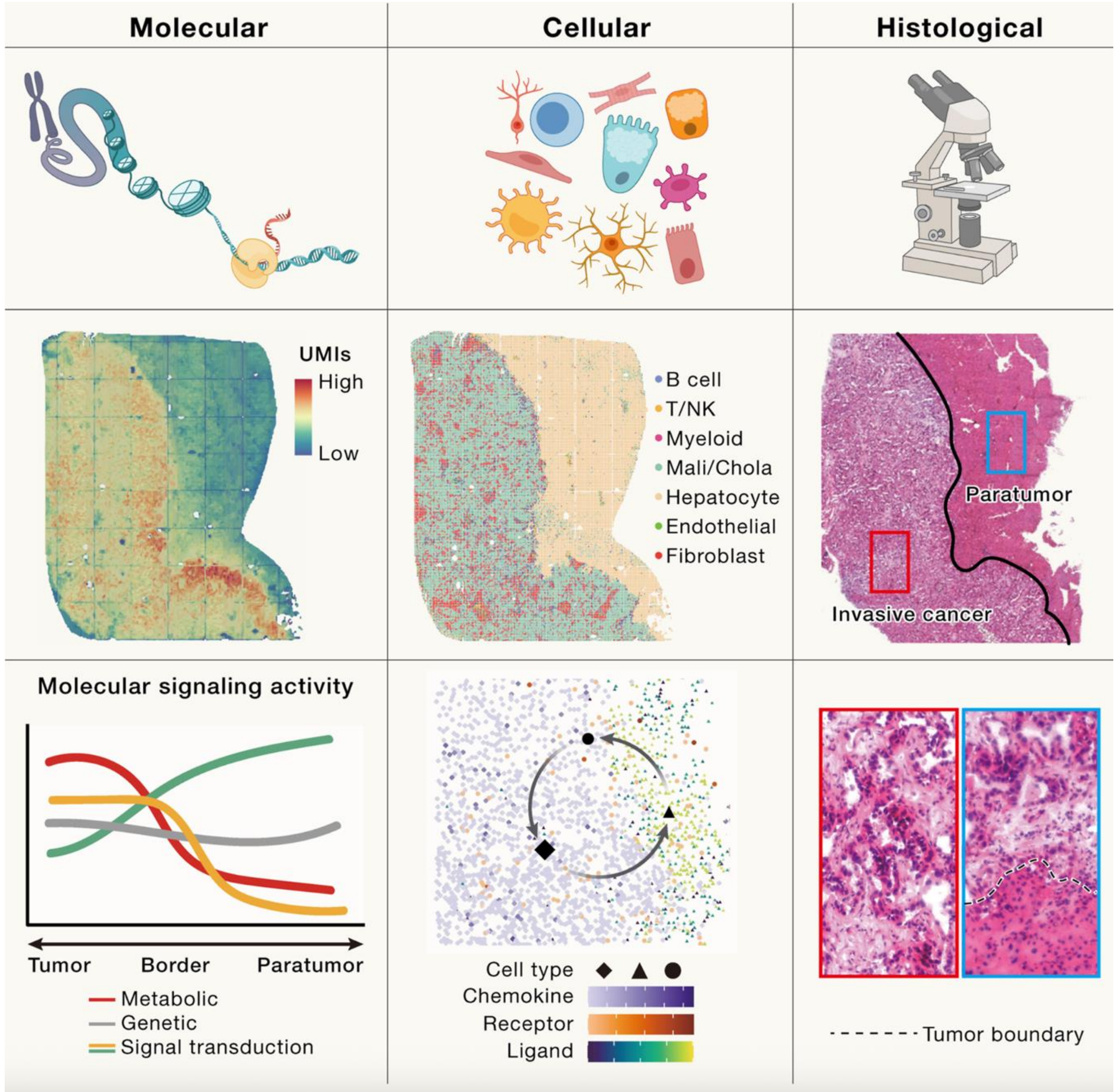
Crash course to spatial omics for today



Park J.. *et al. Genome Biology* 2022.

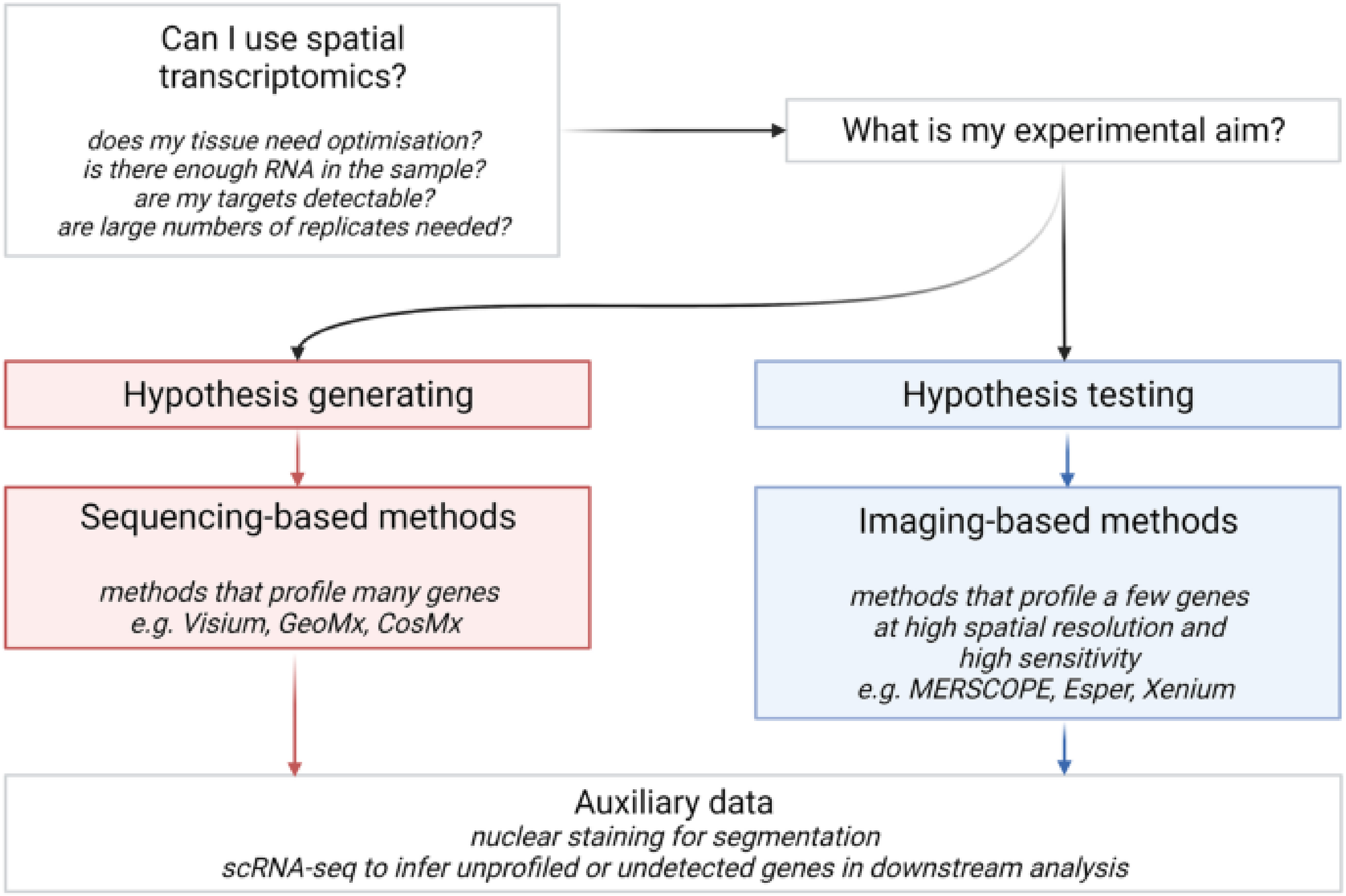
If you want to run spatial omics…

Question: Why do you need spatial omics for your project?



Liu, Longqi et al. *Cell* 2024.

Before starting your spatial journey



Williams, C.G., Lee, H.J., Asatsuma, T. *et al. Genome Med* 2022.

Spatial omics workflow, simplified

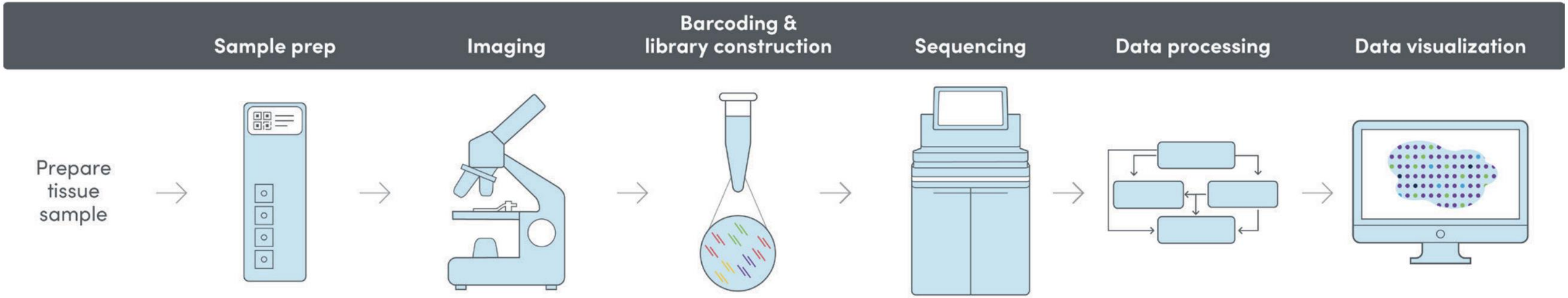


Image from 10X Genomics

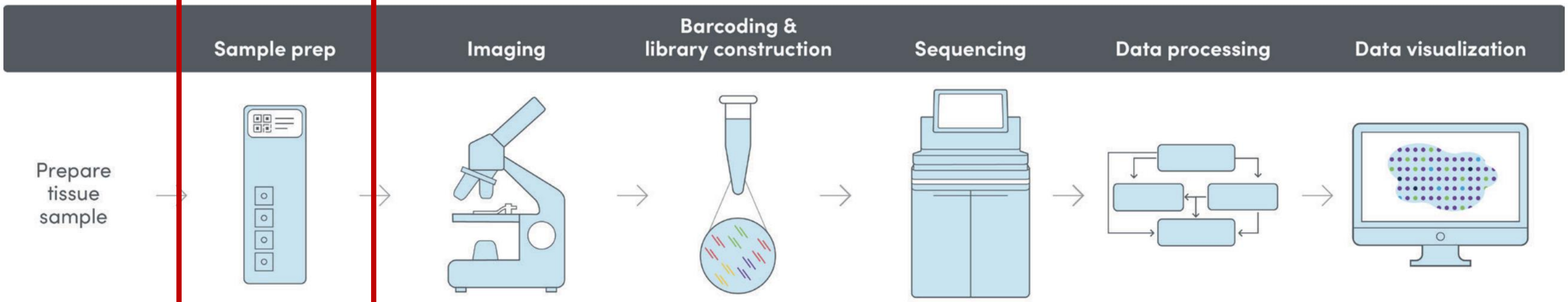
Factors to consider every step

Why Spatial
Cost

Throughput
Capture area
Resolution

Custom panel
Modality
Plex

Controls
Sensitivity, Specificity
Depth

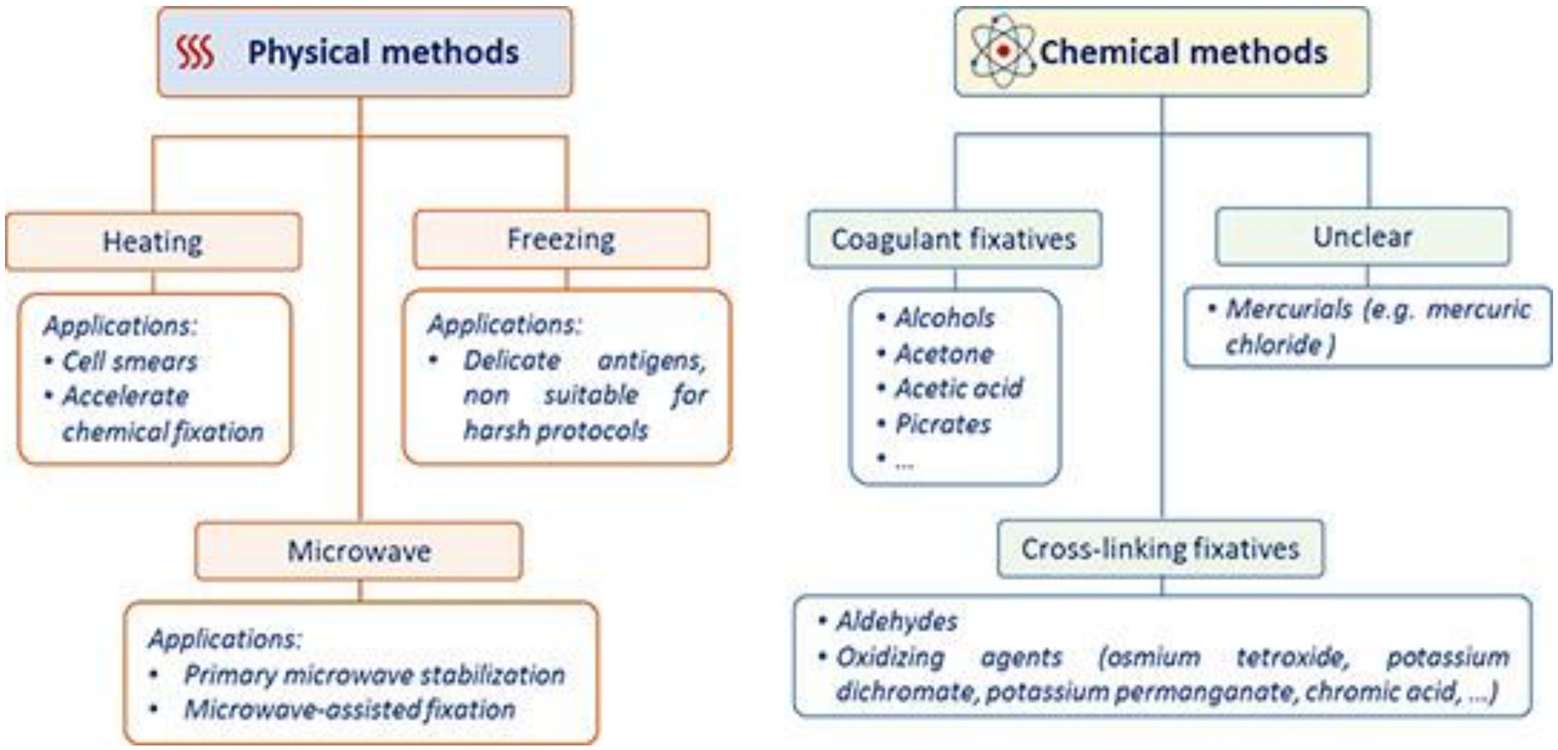


FFPE/FF
Size
Tissue type
Structure vs. RNA

Segmentation
Annotation
Statistics

Image from 10X Genomics

Considerations during sample preparation: fixation



Different protocols but check:

- Uniform penetration, Temperature,
- Fixative concentration & time

pH matters!

Image from Enzo Biosciences

Considerations during sample preparation: preservation

Table 1	FFPE Tissue	Frozen sections
Tissue morphology preservation	V *Yes*	X *Poor*
Long term storage	V *Several years at RT*	X *1 year at -80°C*
Section thickness	V *Thin, 1-50 µm*	X *Thicker → lower resolution*
Enzymes activity and labile antigens preservation	X *No*	V *Yes*
Antigen immunoreactivity	X *Epitope masking, antigen retrieval often necessary*	V *No epitope masking*
Fixation	V *Prior to embedding*	V *After sectioning*

- Antigen retrieval protocol may vary
- Decalcification protocol
- Tissue clearing for some platforms
- Some tissues tend to be more sensitive to degradation (enzymatic activities)

Factors to consider every step

Why Spatial

Cost

Throughput

Capture area

Resolution

Custom panel

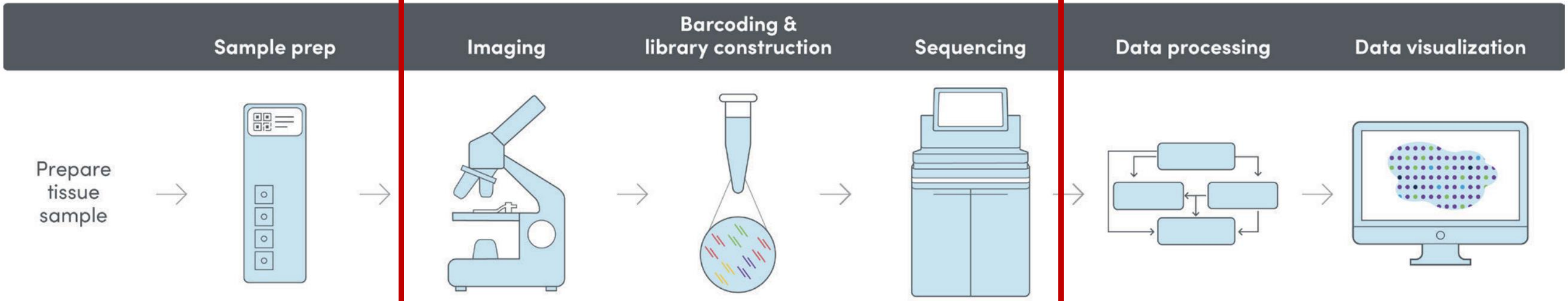
Modality

Plex

Controls

Sensitivity, Specificity

Depth



FFPE/FF

Size

Tissue type

Structure vs. RNA

Segmentation

Annotation

Statistics

Image from 10X Genomics

Available platforms to consider

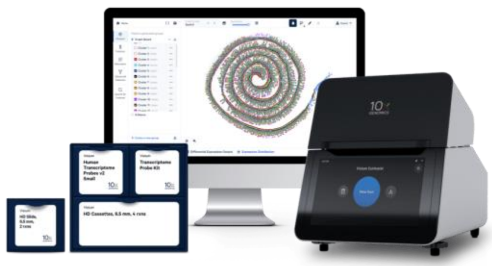
CosMx (EIPM)

IMC (EIPM)

CellScape

CODEX (Phenocycler)

Xenium



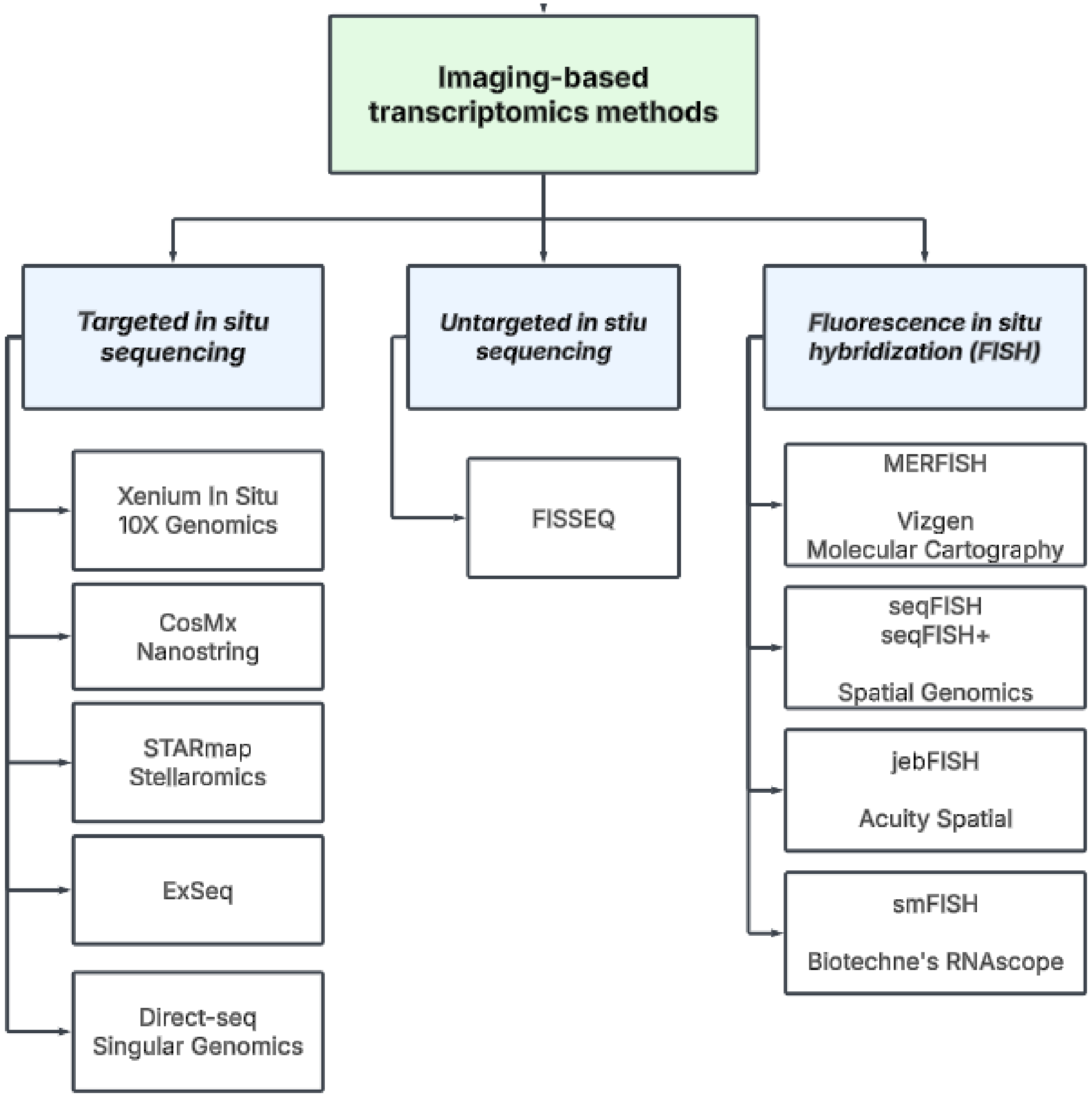
VisiumSD/HD

Stereo-seq

Questions?

Imaging-based spatial transcriptomics

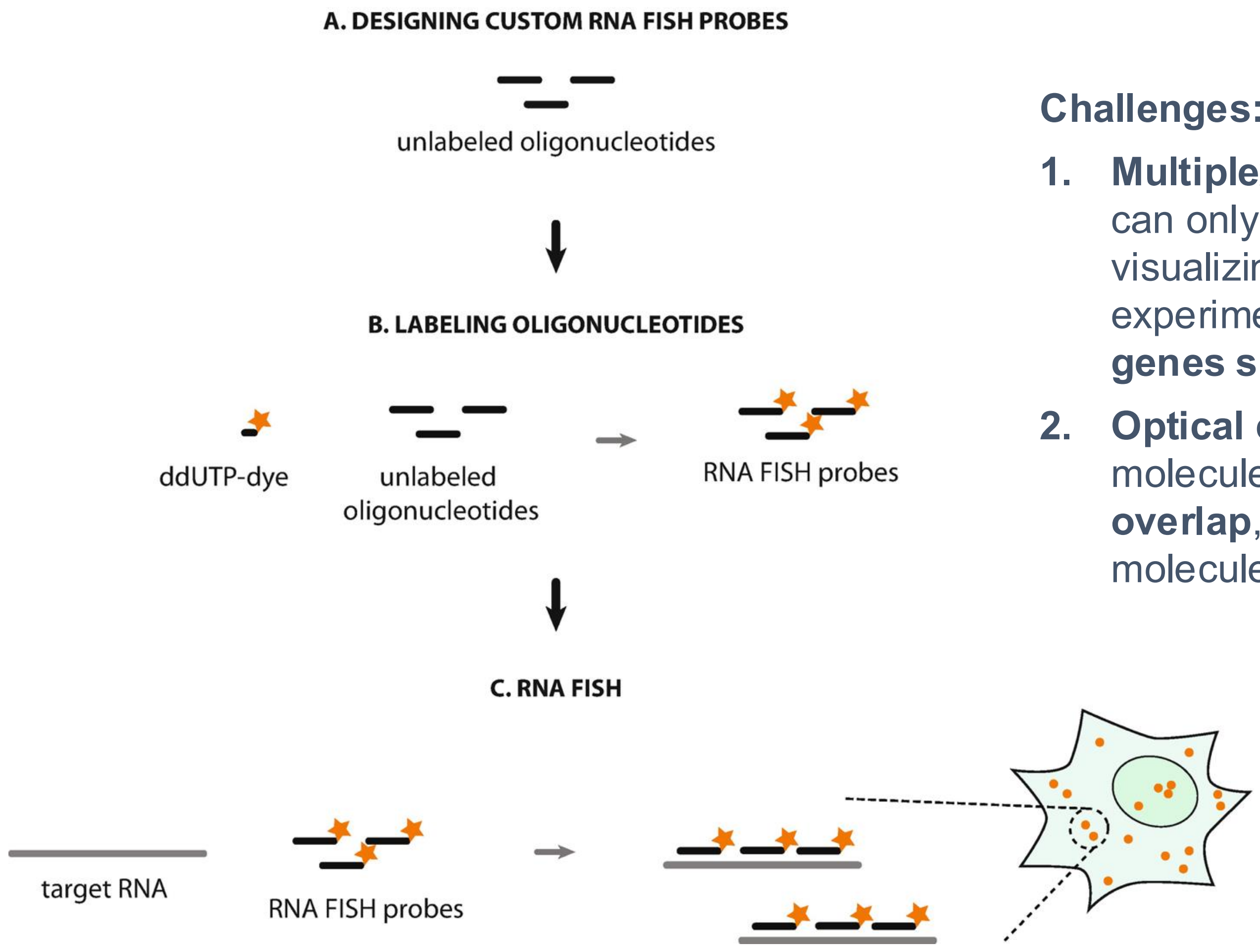
Imaging based spatial transcriptomics



- *In situ* **sequencing (ISS)** involves hybridizing probes to target transcripts, amplifying the signal through techniques like rolling circle amplification, and then sequentially reading nucleotide sequences using iterative rounds of fluorescent labeling and imaging.
- *In situ* **hybridization (ISH)**, detects specific RNA or DNA sequences by hybridizing many fluorescently labeled probes directly to target molecules, enabling spatial localization.

Diagram from LatchBio

Basic concept for fluorescence in situ hybridization (FISH)

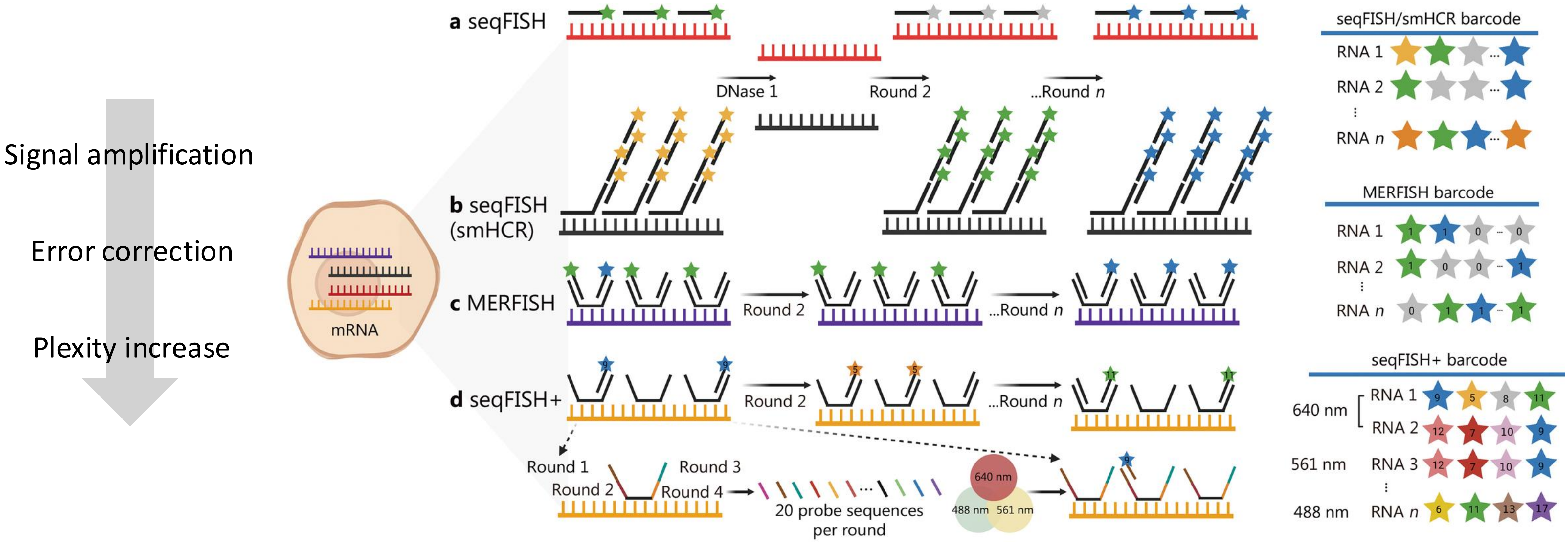


Challenges:

1. **Multiplexing constraints:** Traditional fluorescence microscopes can only detect a few fluorophores at once, limiting researchers to visualizing only a **handful of RNA species** in a single experiment. What if scientists wanted to track **thousands of genes simultaneously**?
2. **Optical crowding:** In densely packed cellular environments, RNA molecules can cluster so tightly that **fluorescence signals overlap**, making it difficult to distinguish and count individual molecules.

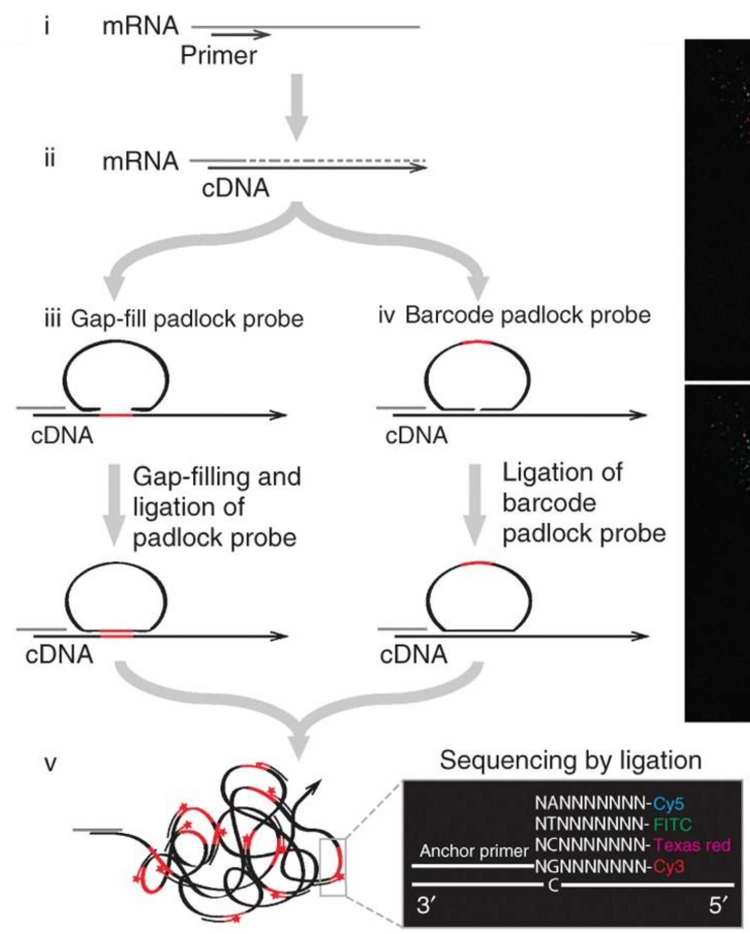
Piskadlo E *et al. Springer Protocols: Alternative Splicing* 2022.

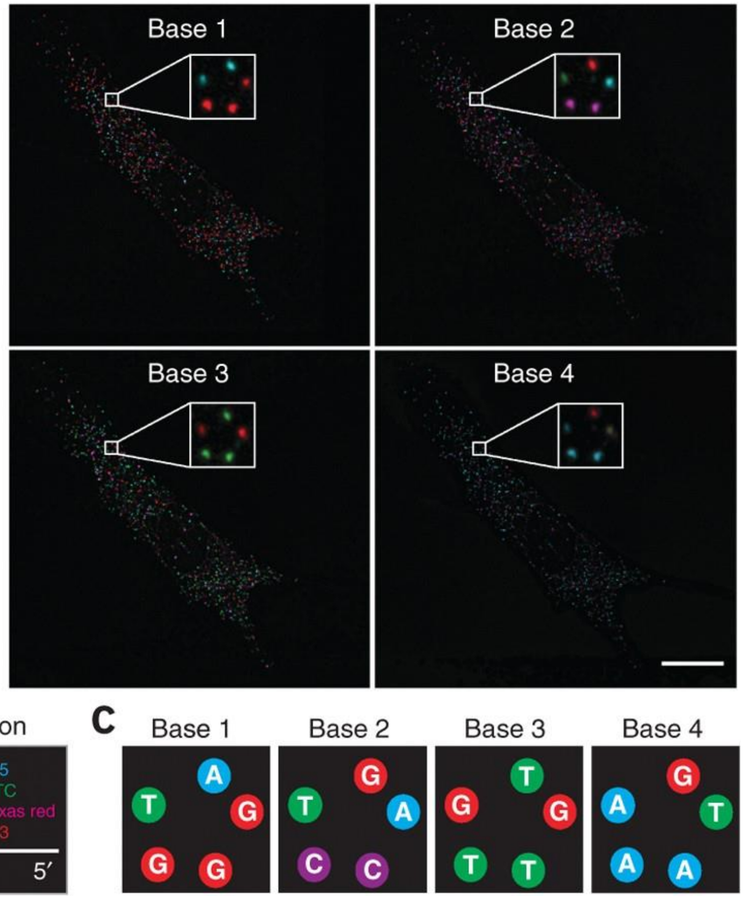
Barcoding and detection approach vary by technology

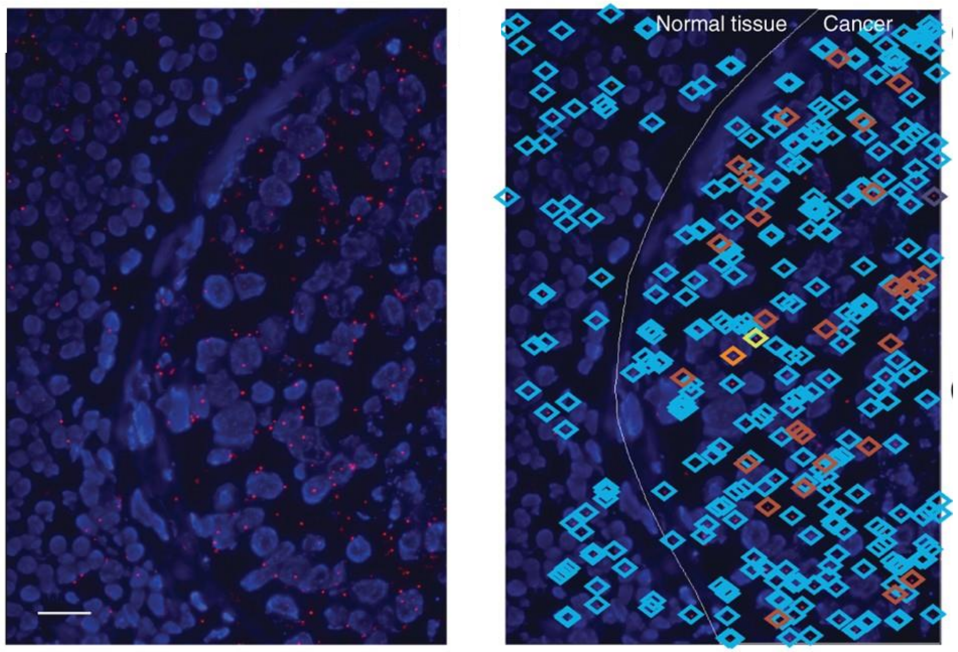


Wang, WJ., Chu, LX., He, LY. *et al. Military Med Res* 2023.

In Situ Sequencing (ISS)





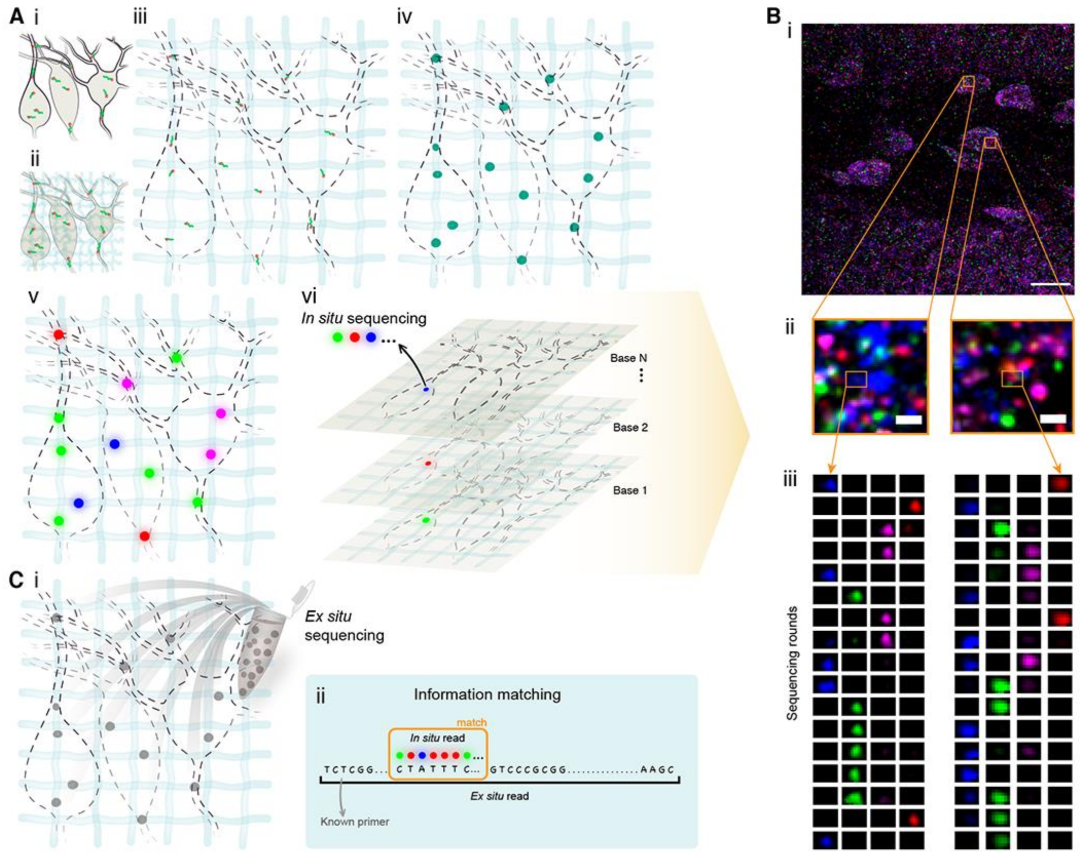


low gene detection efficiency
(low efficiency of *in situ* cDNA synthesis)

Ke, R., Mignardi, M., Pacureanu, A. *et al. Nature Methods* 2013.

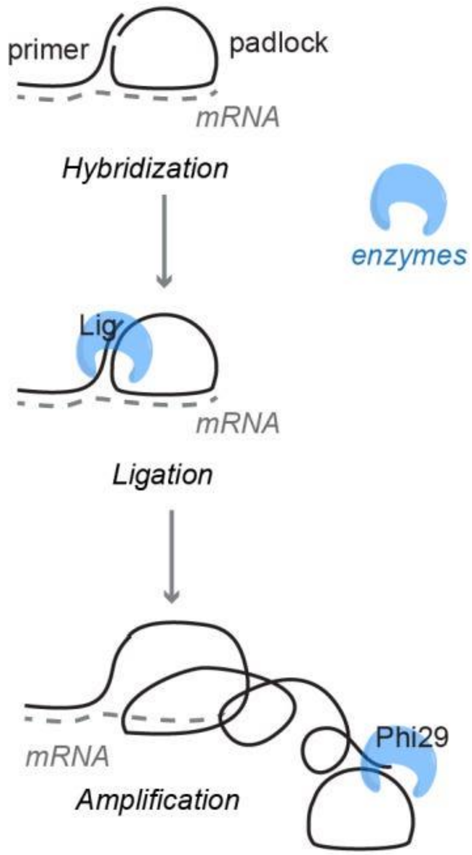
Improvements in the workflow

Expansion sequencing (ExSeq)



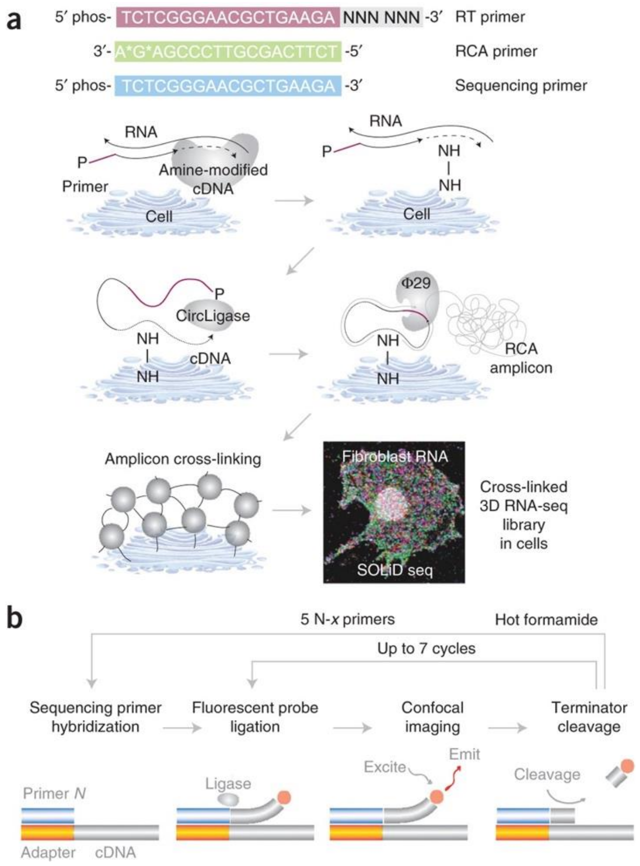
Hydrogel to solve optical crowding (Chemical anchor and swelling the tissue) -> increased detection efficiency.

STARmap – SNAIL probe



No RT, better capture efficiency
Later STARmap PLUS & RIBOmap with various probe design optimization efforts.

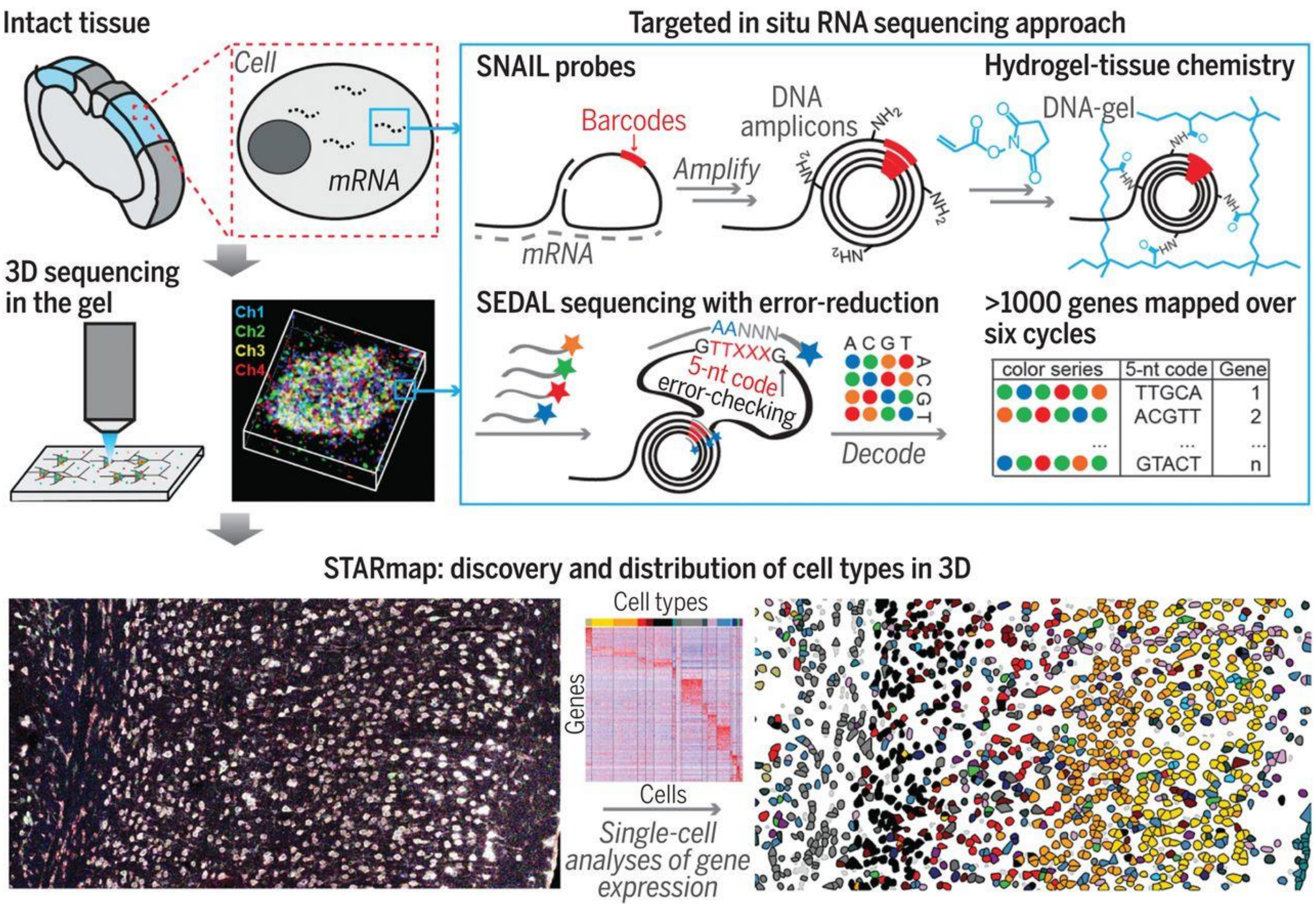
FISSEQ

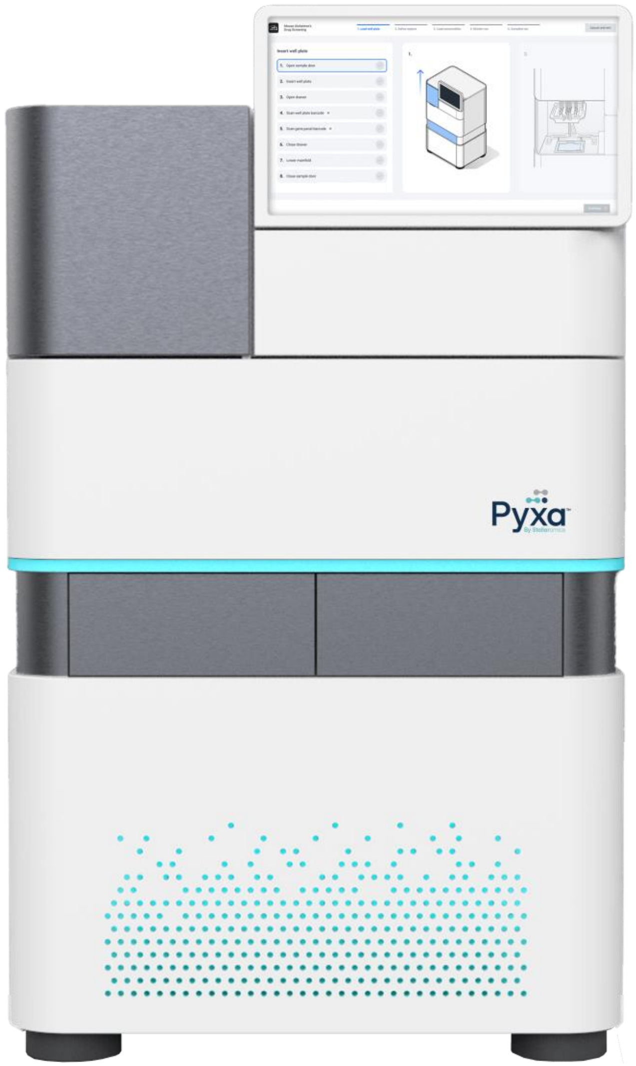


Sequencing by ligation; untargeted, high multiplexing capacity but low efficiency.

Ke, R., Mignardi, M., Pacureanu, A. *et al. Nature Methods* 2013.; Xiao Wang et al. *Science* 2018.; Lee, J., Daugharthy, E., Scheiman, J. et al. *Nature Protocols* 2015.

Going 3D – coming soon

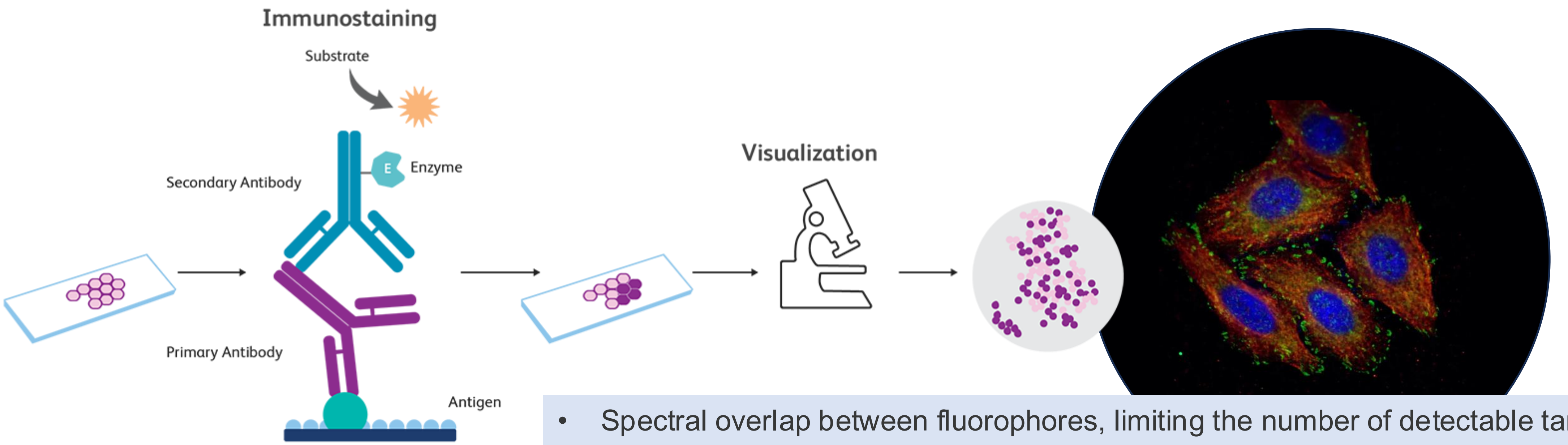




•Xiao Wang *et al*. *Science* 2018.

Imaging-based spatial proteomics

Immunostaining to detect protein in the lab

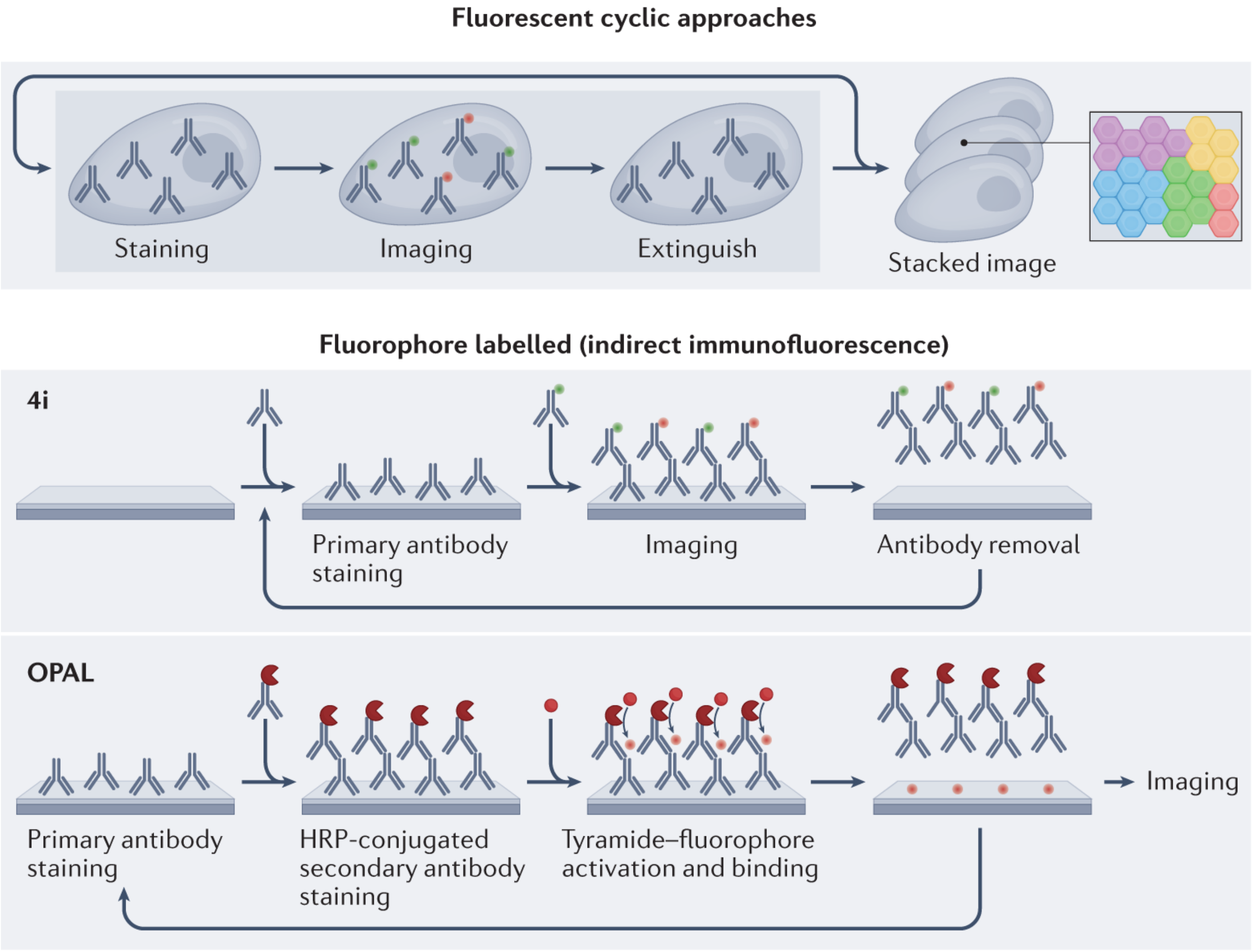


- Spectral overlap between fluorophores, limiting the number of detectable targets
- Cross-reactivity of antibodies, leading to non-specific staining
- Limited multiplexing capacity, restricting simultaneous detection of multiple markers
- Tissue autofluorescence, reducing signal clarity in fluorescence-based IHC

Immediate solution: remove the tags after use

Image from BD

Cyclic Multiplexed Immunofluorescence Methods

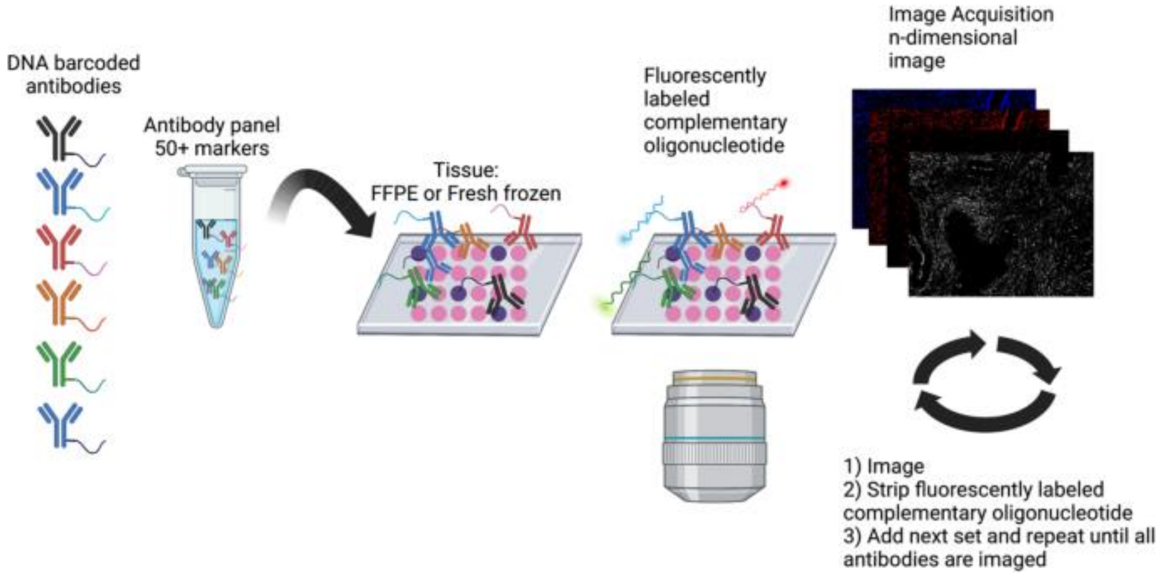


Signal removal with:

- fluorophore inactivation by oxidation (e.g., hydrogen peroxide)
- Alkaline pH
- Chemical bleaching (e.g., LiBH4/lithium borohydride)
- Light (photobleaching)
- Antibody elution buffers

Moffitt, J.R., Lundberg, E. & Heyn, H. *Nat Rev Genet* 2022.

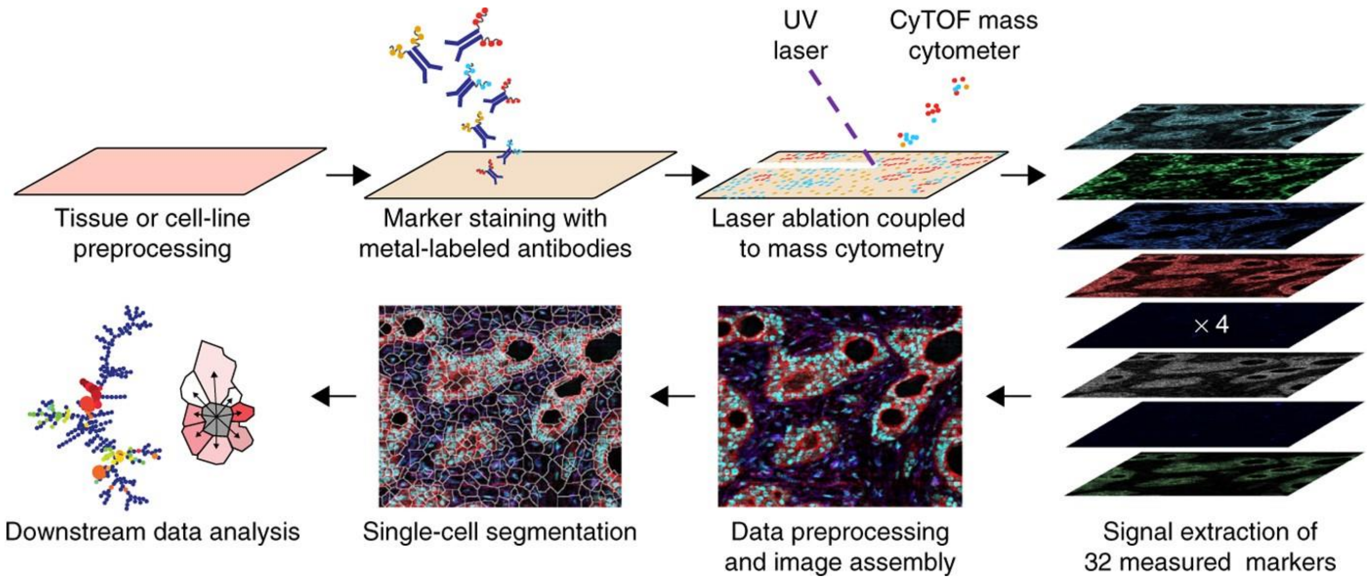
Adding DNA reporter tags to improve detection



- After this initial staining, the detection process uses multiple rounds of imaging without additional antibody applications.
- Each imaging cycle, fluorescently labeled reporter probes complementary to specific DNA tags hybridize to their target sequences, generating detectable signals.
- When these hybridization events occur, the resulting fluorescent patterns serve as unique identifiers for different proteins.
- Between imaging cycles, the reporter probes are removed through rapid chemical or enzymatic processes, allowing new probes to bind in subsequent rounds.

Image from CODEX

Mass Cytometry readout instead of imaging

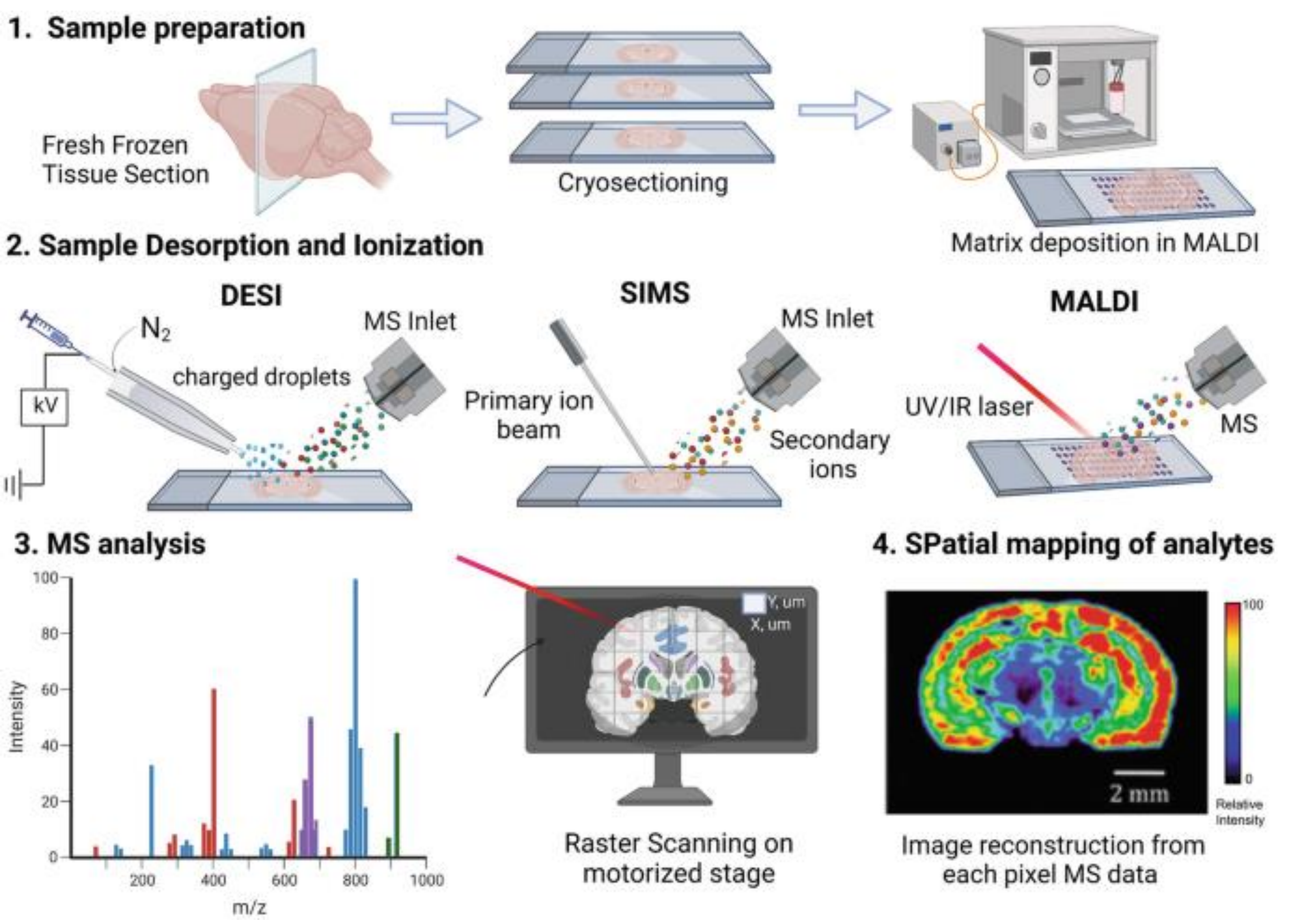


- **High Multiplexing:** protein targets (up to 40 or more) in a single tissue section.
- **Preservation of Tissue Architecture:** IMC preserves the spatial organization and relationships of cells within the tissue, unlike flow cytometry which requires cell dissociation.

- **Lower Resolution:** relatively lower (1 um) compared to other proteomics imaging techniques
- **Limited Availability of Isotopes:** The number of available isotopes for staining can be a limiting factor in expanding the number of markers that can be simultaneously analyzed. In some cases, metal contamination from therapeutic or diagnostic agents can interfere with IMC signal detection.
- **Cost:** Reagents (e.g., metal-tagged antibodies) can be expensive, and difficult to optimize (and harmful for the environment)

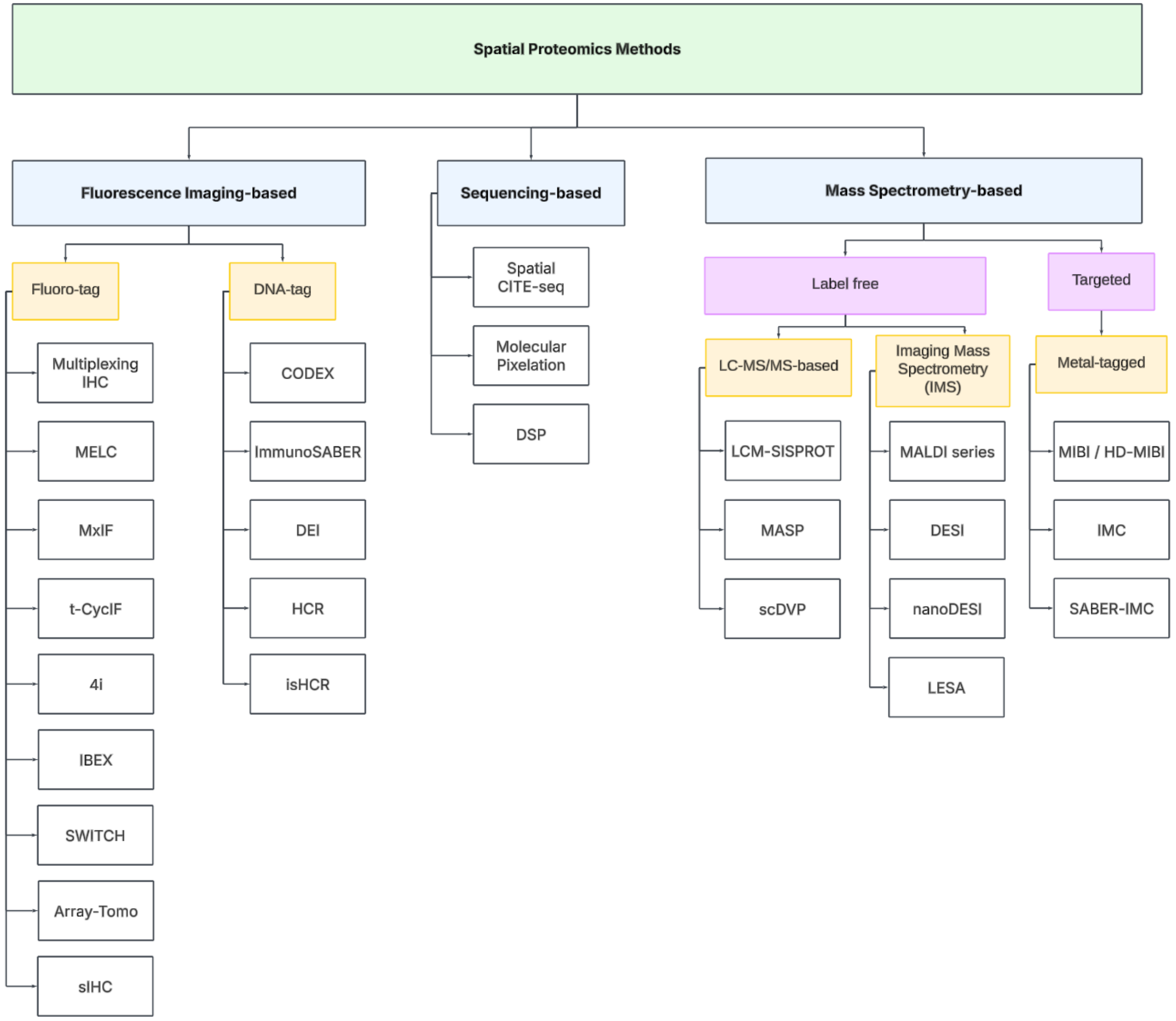
Giesen, C., Wang, H., Schapiro, D. *et al. Nature Methods* 2014.

Same concept for metabolomics



Vijaya Lakshmi, K. Metabolomics. Springer, Cham. 2023.

Spatial proteomics development for better detection



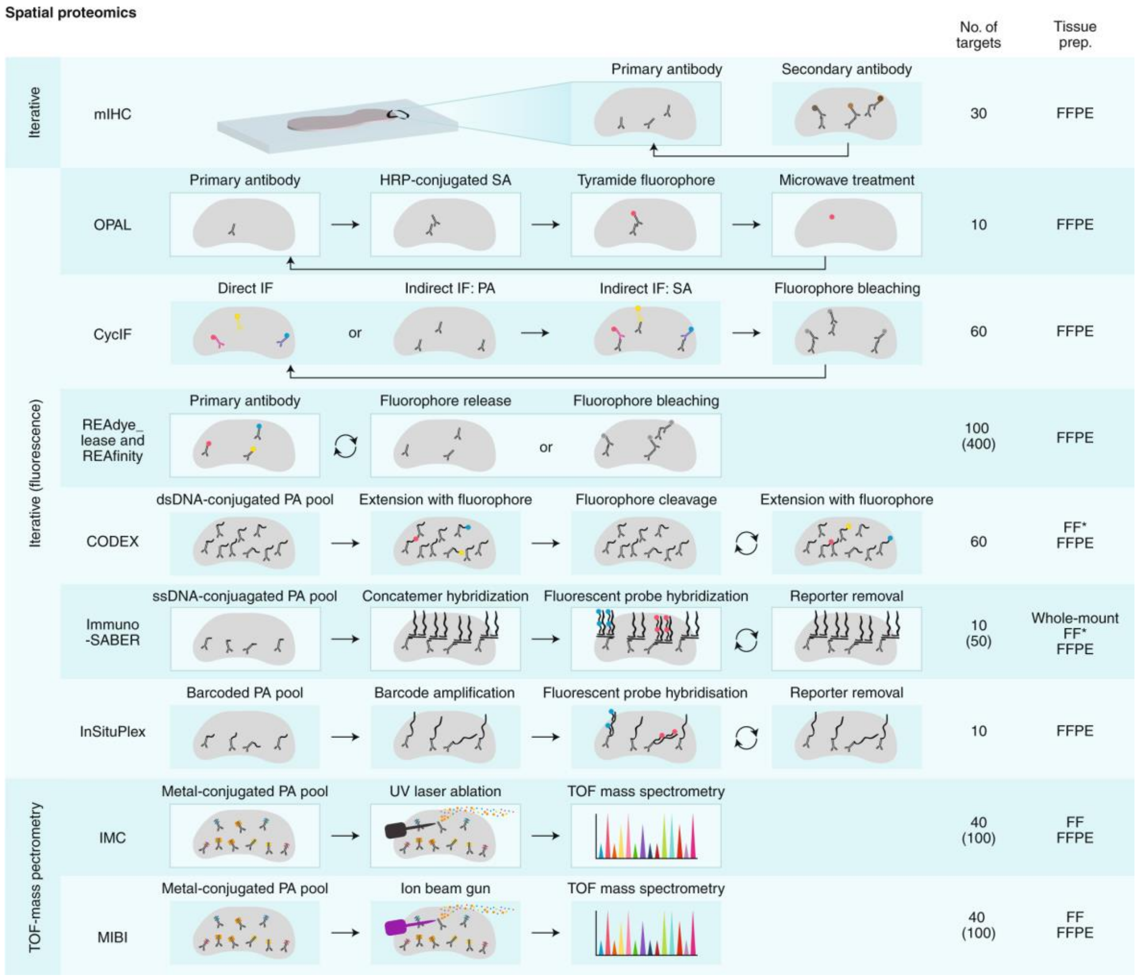


Diagram from LatchBio; Lewis, S.M., Asselin-Labat, ML., Nguyen, Q. *et al.* Nature Methods 2021.

Going for multi-omics

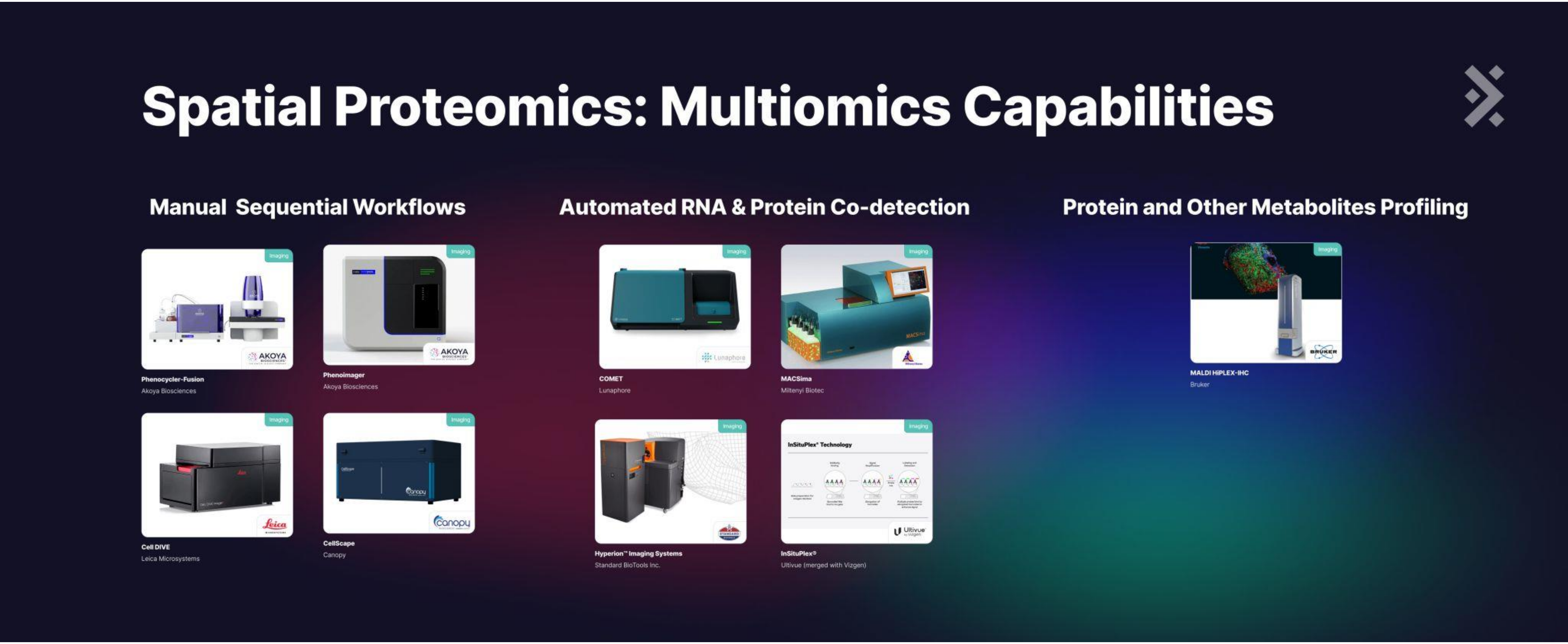
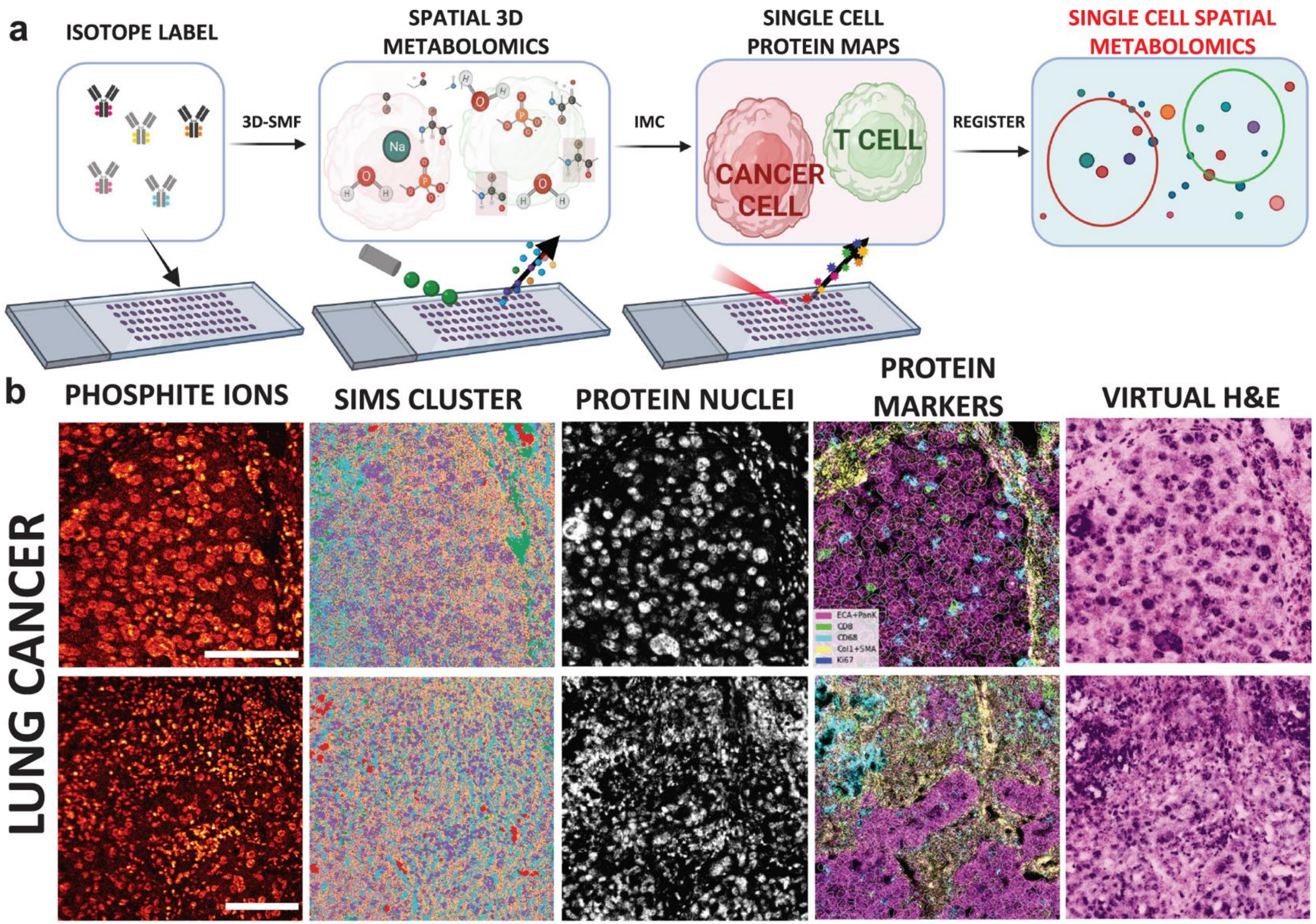


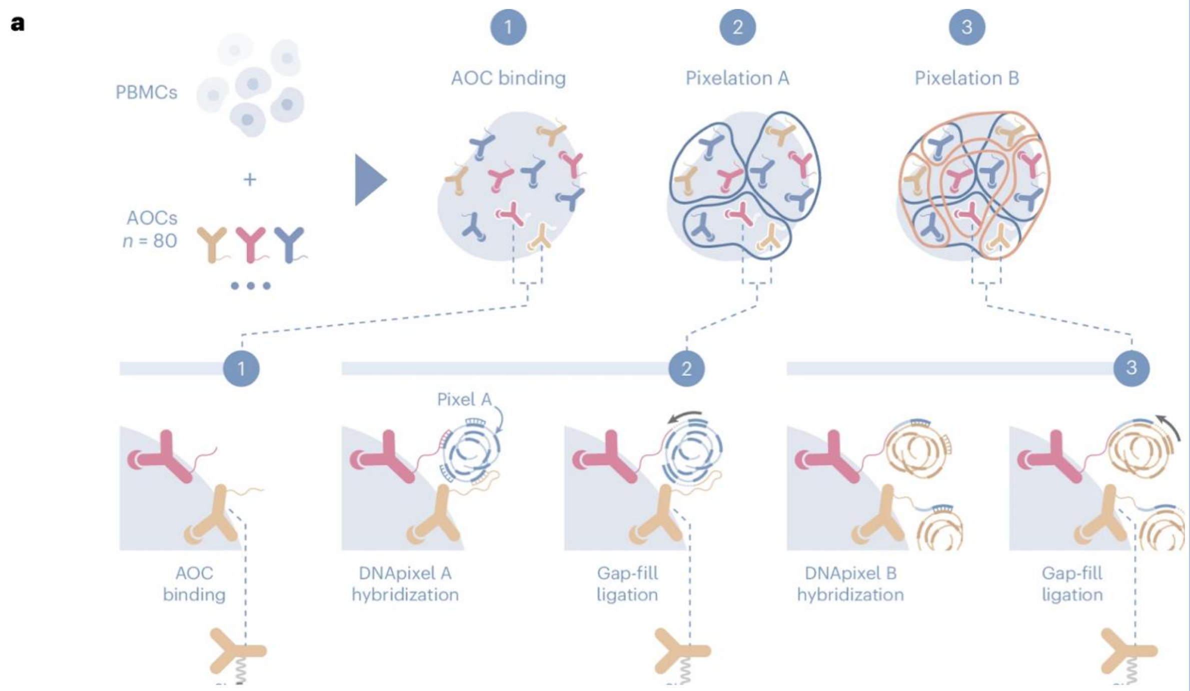
Image from LatchBio

Spatial metabolomics + proteomics



Hu, T., Allam, M., Cai, S. *et al. Nature Communications* 2023.

And beyond traditional proteomics: Molecular pixelation



1. **Protein abundance:** Similar to flow cytometry or CyTOF, measuring the overall expression levels of proteins per cell.
2. **Individual protein spatial distribution:** Quantifying whether a single protein is randomly distributed, polarized, or clustered within the cell.
   - Polarity scores indicate whether a protein is organized (<0), clustered (>0), or randomly dispersed (=0).
   - Example: T cells stimulated via CD3 show polarized CD3 localization on one end of the cell.
3. **Protein colocalization analysis:** Determining how different proteins are positioned relative to each other on the same cell.
   1. Colocalization scores range from <0 (segregated proteins) to >0 (proteins physically co-localized).
   2. Example: Rituximab treatment of RAJI cancer cells polarizes CD20, a mechanism critical for CAR-T therapy and antibody-based drugs.

Karlsson, F., Kallas, T., Thiagarajan, D. *et al. Nature Methods* 2024.

Sequencing-based spatial transcriptomics

Sequencing based spatial transcriptomics

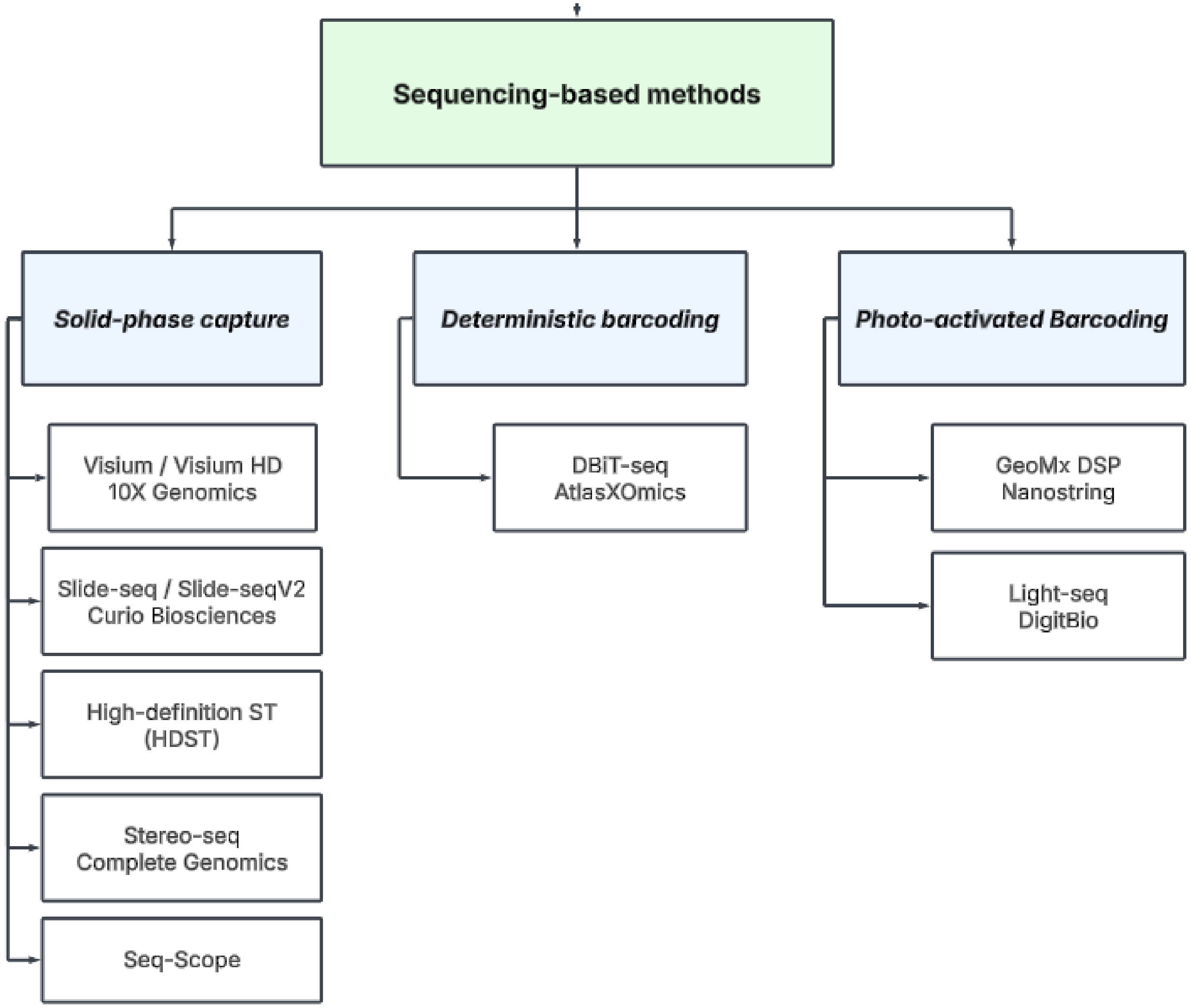
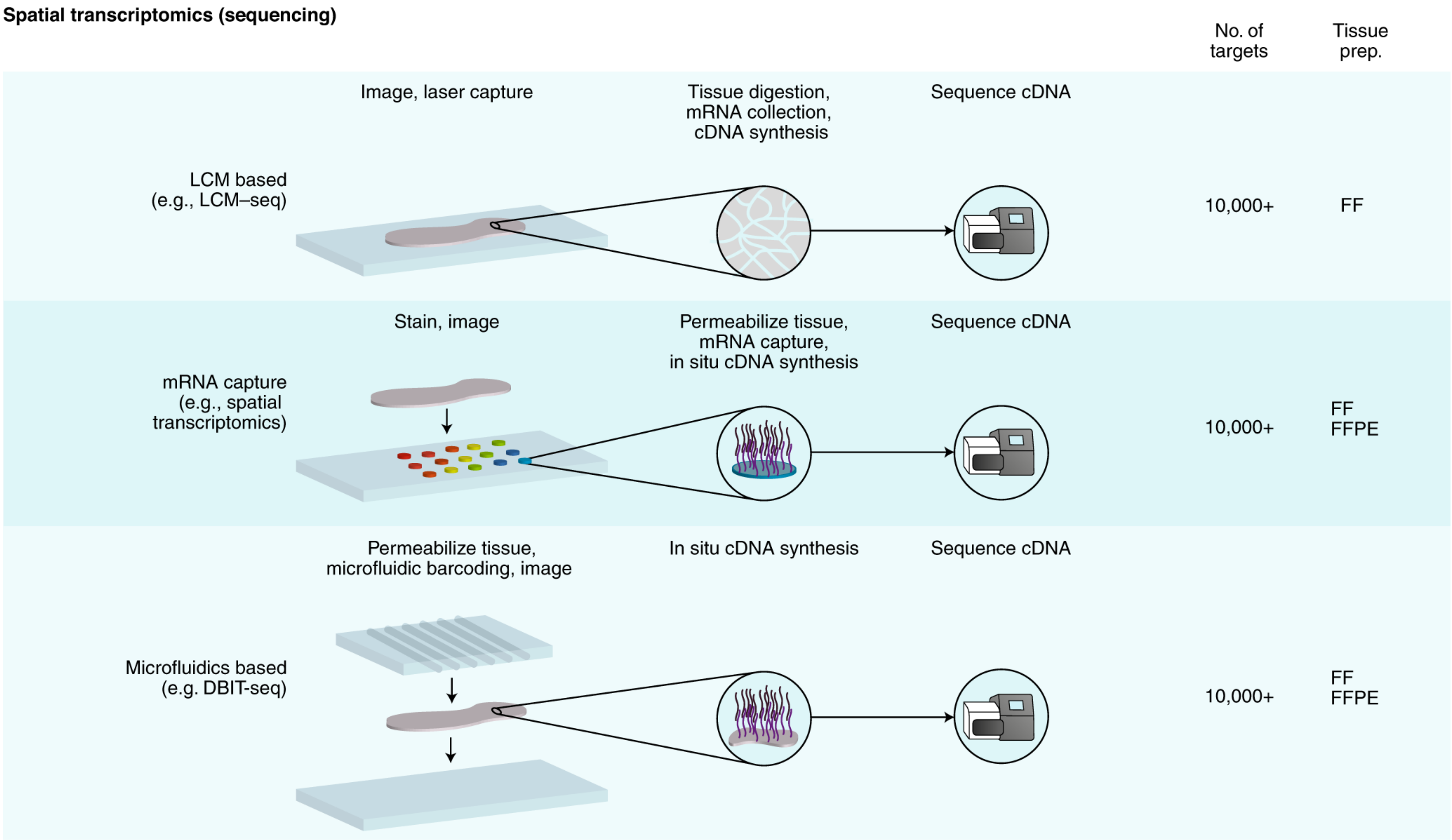


Diagram from LatchBio

- Method that ultimately uses sequencing to decode spatial information: Involves library preparation & sequencing
- Spatial information is in the barcodes, molecular information is in the sequencing library

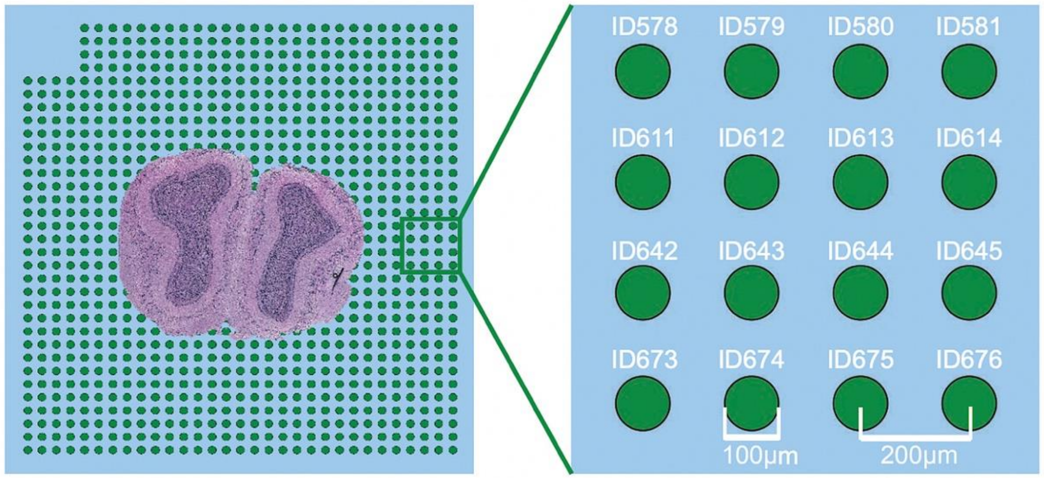
- Varied by **capture methods (spatial indexing and/or ligation of spatial barcodes)**
  - Capture on barcoded surface
  - Applying barcode via device
  - Apply barcode directly on tissue

Capture method varies within sequencing-based spatial omics



Lewis, S.M., Asselin-Labat, ML., Nguyen, Q. *et al.* Nature Methods 2021.

First version of spatial sequencing: 2016, Ståhl et al.



- **~200 million oligonucleotides across 1007 features**
- **100 mm** diameter**,** 200 mm center-to-center distance
- capture area of 6.2 X 6.6 mm^2 (**100-200 µm** resolution)

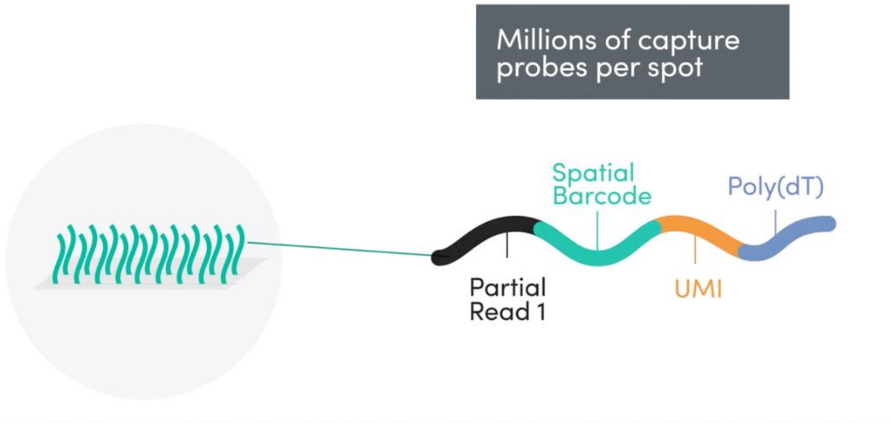
Capture rate were much lower than mFISH methods

Each Oligonucleotide on barcoded slide contained:

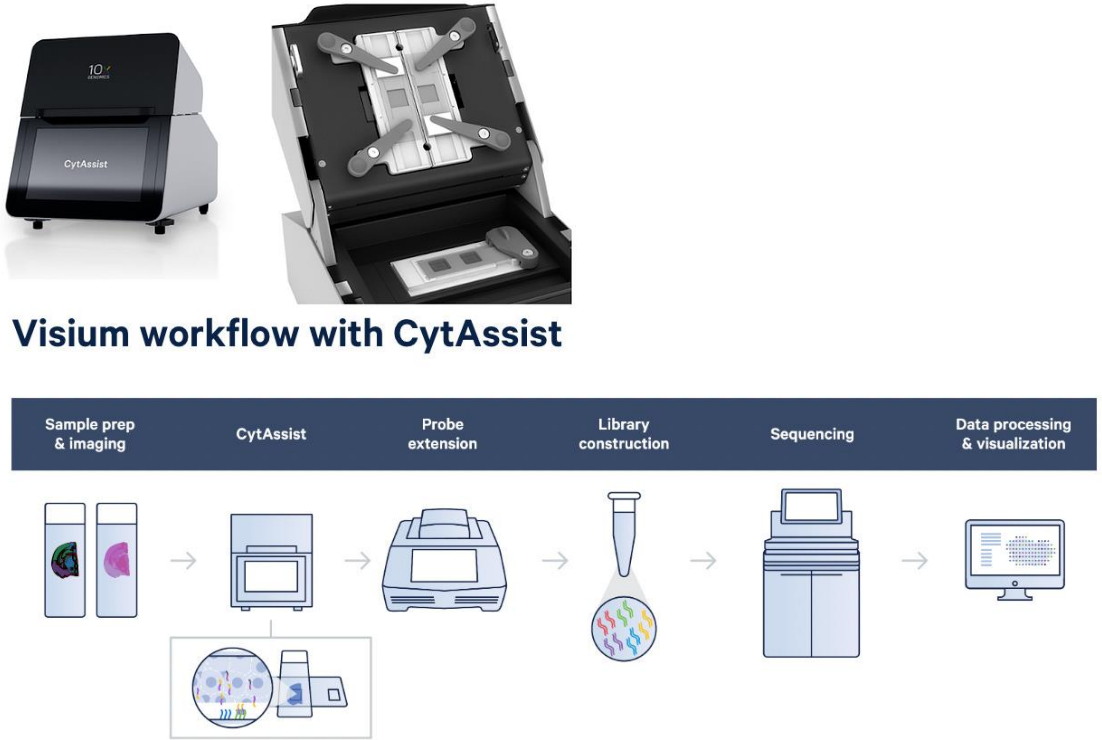
1. A **cleavage site** to release captured molecules during processing,
2. An **amplification and sequencing handle** for downstream library preparation,
3. A **spatial barcode** encoding positional information,
4. A **unique molecular identifier (UMI)** to track individual transcripts and minimize PCR biases, and
5. An **mRNA capture region** with a poly(dT) tail to hybridize polyadenylated mRNA from the tissue.

Image from LatchBio

Visium (2019), first commercialization of the technology



- resolution that is **2-4X higher**: The slide contains approximately **5,000 spots**, each with a **55 µm** diameter
- Offering FFPE support, streamline with CytAssist (2022)

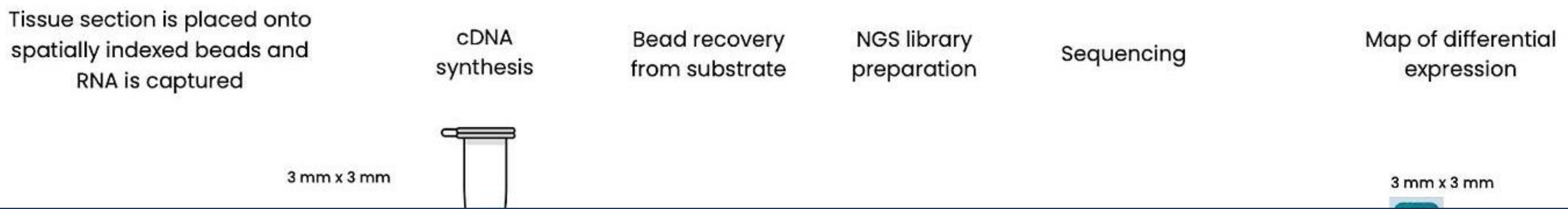


Facilitate transfer of transcriptomic probes in FFPE or fresh frozen samples with Visium CytAssist. In the Visium CytAssist workflow, sectioning, tissue preparation, staining (H&E or IF), and imaging take place on a standard glass slide. After probe hybridization, two standard glass slides and a Visium slide with two Capture Areas are placed in the CytAssist instrument so that the tissue sections on the standard slides can be aligned on top of the Capture Areas. Within the instrument, a brightfield image is captured to provide spatial orientation for data analysis, followed by hybridization of transcriptomic probes to the Visium slide. The remaining steps, starting with probe extension, follow the standard Visium workflow outside of the instrument.

Image from LatchBio, 10X Genomics

Slide-seq commercialization with Curio for higher resolution

Slide-seq (Rodriques et al.) increased capture efficiency by applying **10 µm** beads onto a thin glass



2024: True single-cell resolution after optimization (Curio Trekker) by directly diffusing barcodes to cell nuclei using UV

+ No instrument needed, simple workflow
- Sequencing cost, only FF

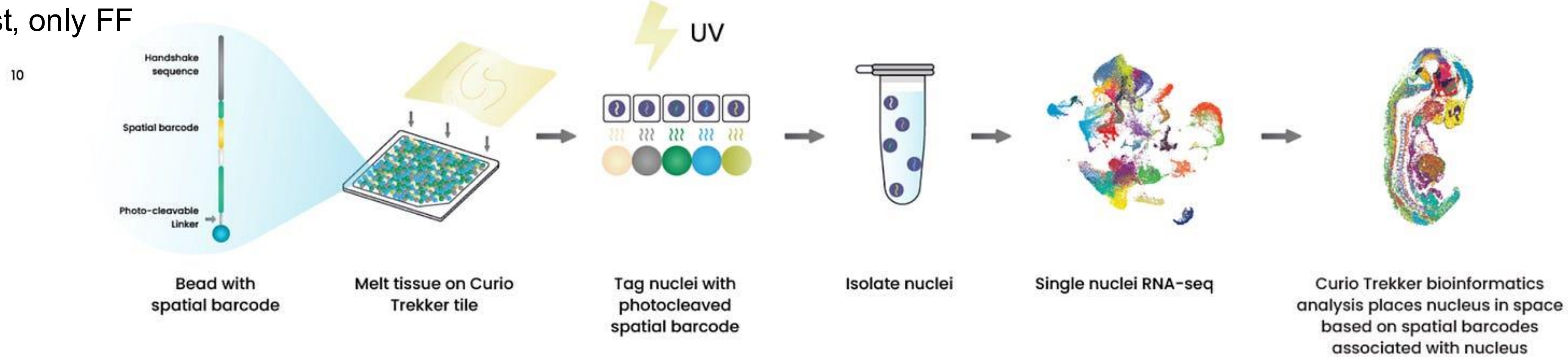
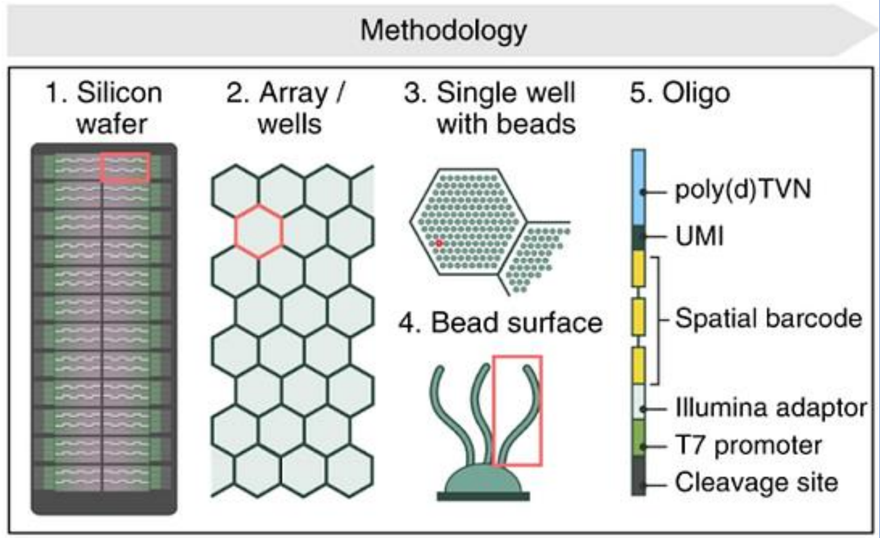
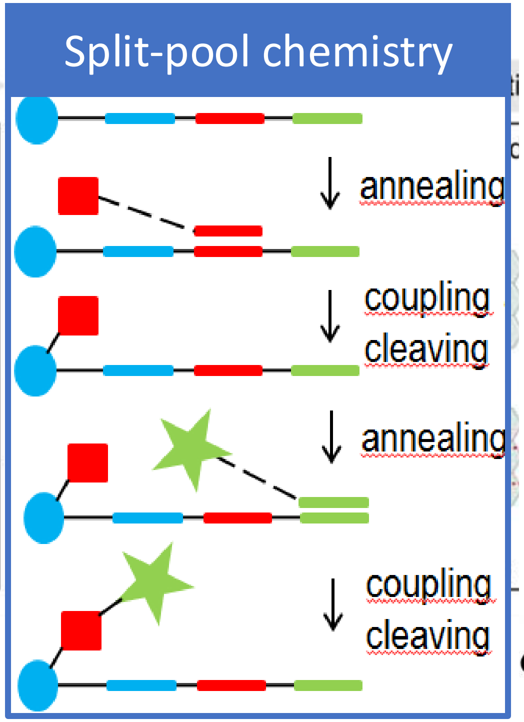
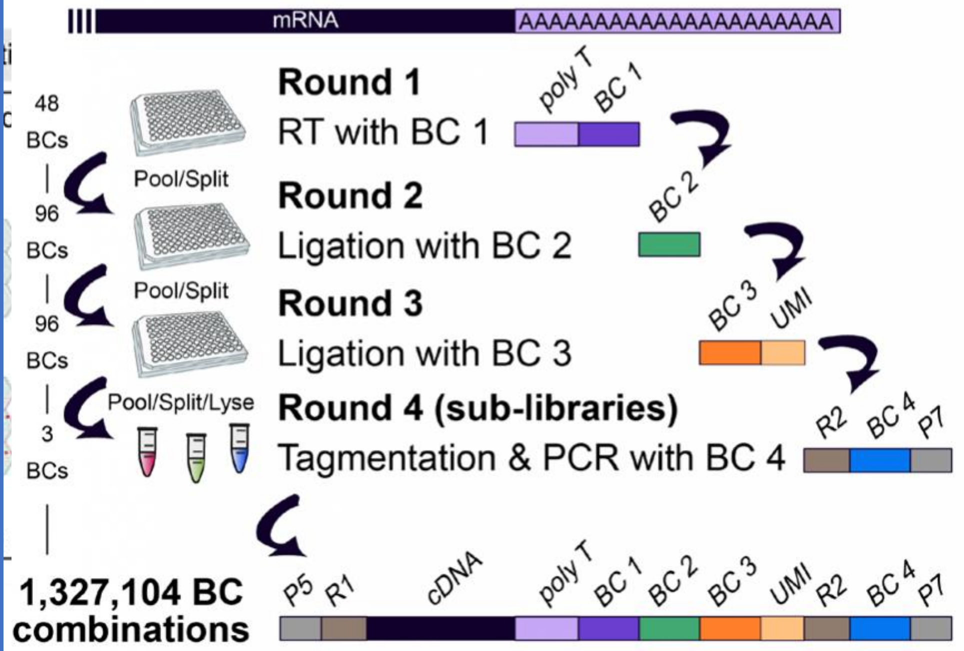


Image from LatchBio, Curio Biosciences

Pushing for high resolution with high capture rate

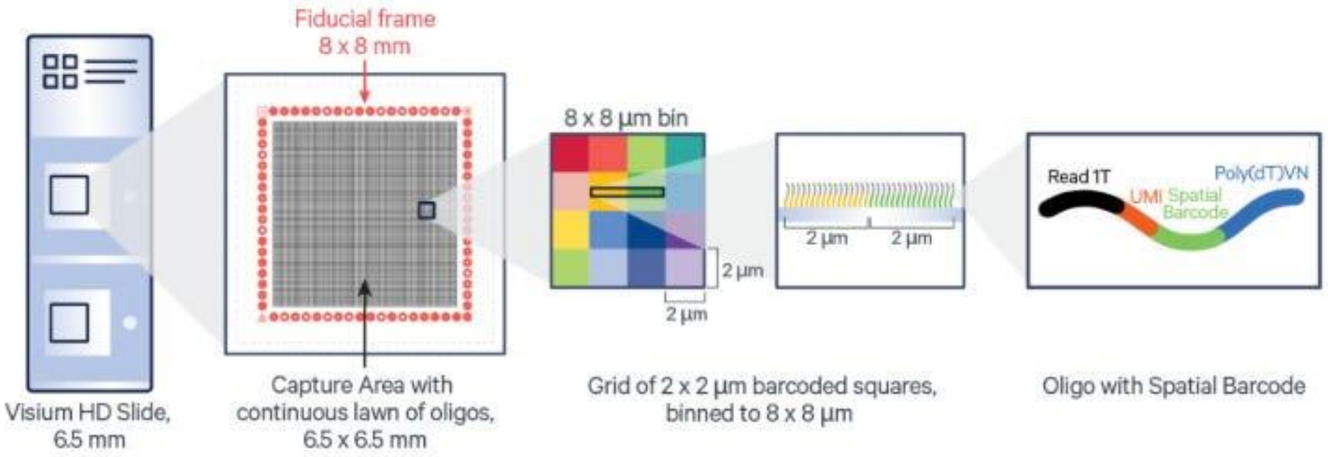




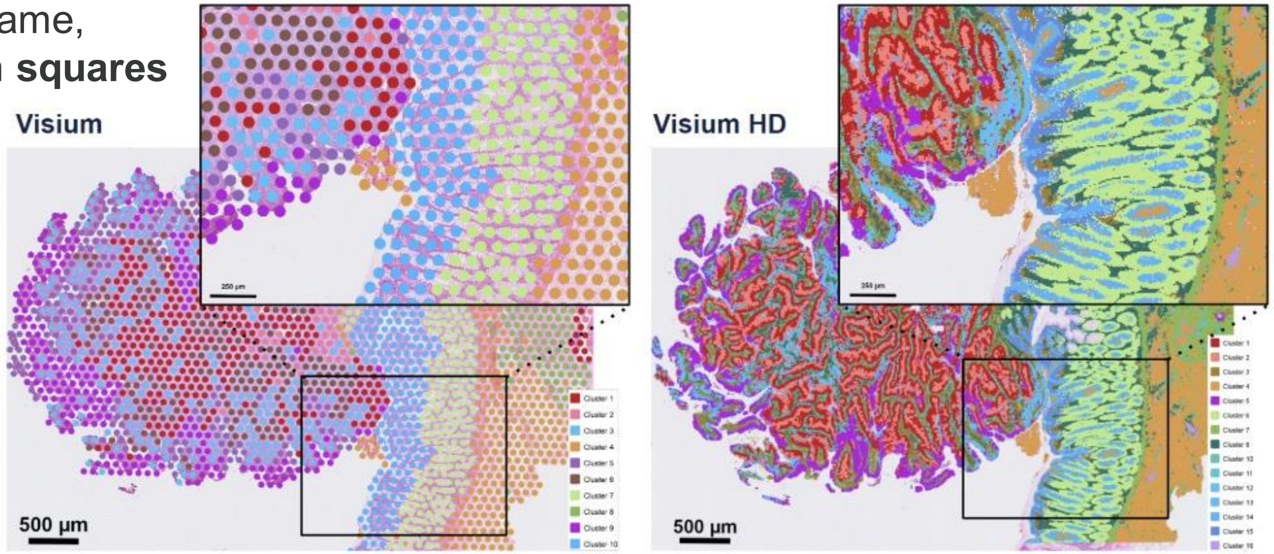


Vickovic, S., Eraslan, G., Salmén, F. *et al. Nature Methods* 2019; Image of split-pool synthesis from Wikipedia and prelights

Similar approach with VisiumHD

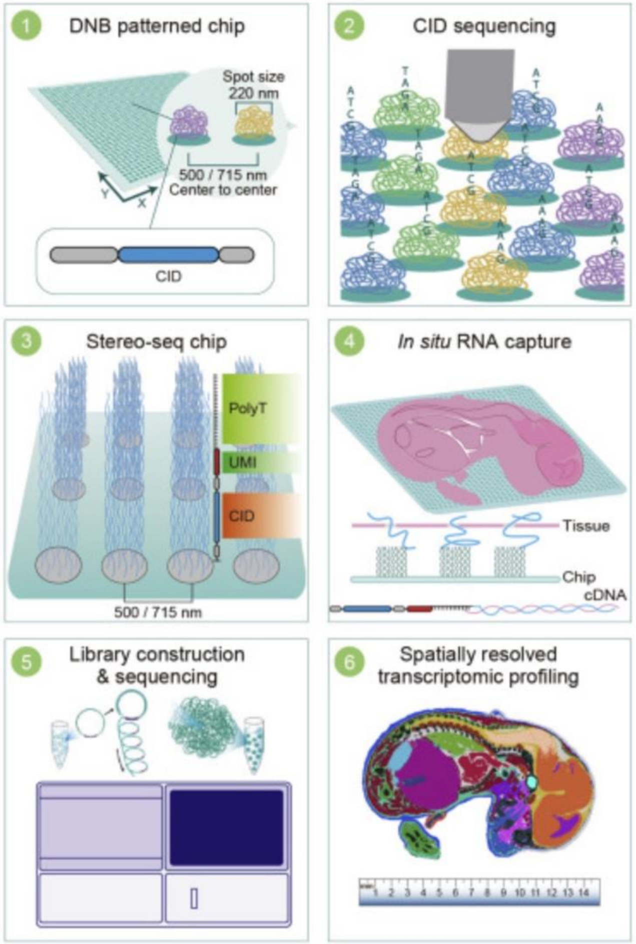


two 6.5 × 6.5 mm capture areas within an 8 × 8 mm fiducial frame, densely packed with **~11 million uniquely barcoded 2 × 2 µm squares** (typically analyzed at 8 x 8 µm bins)

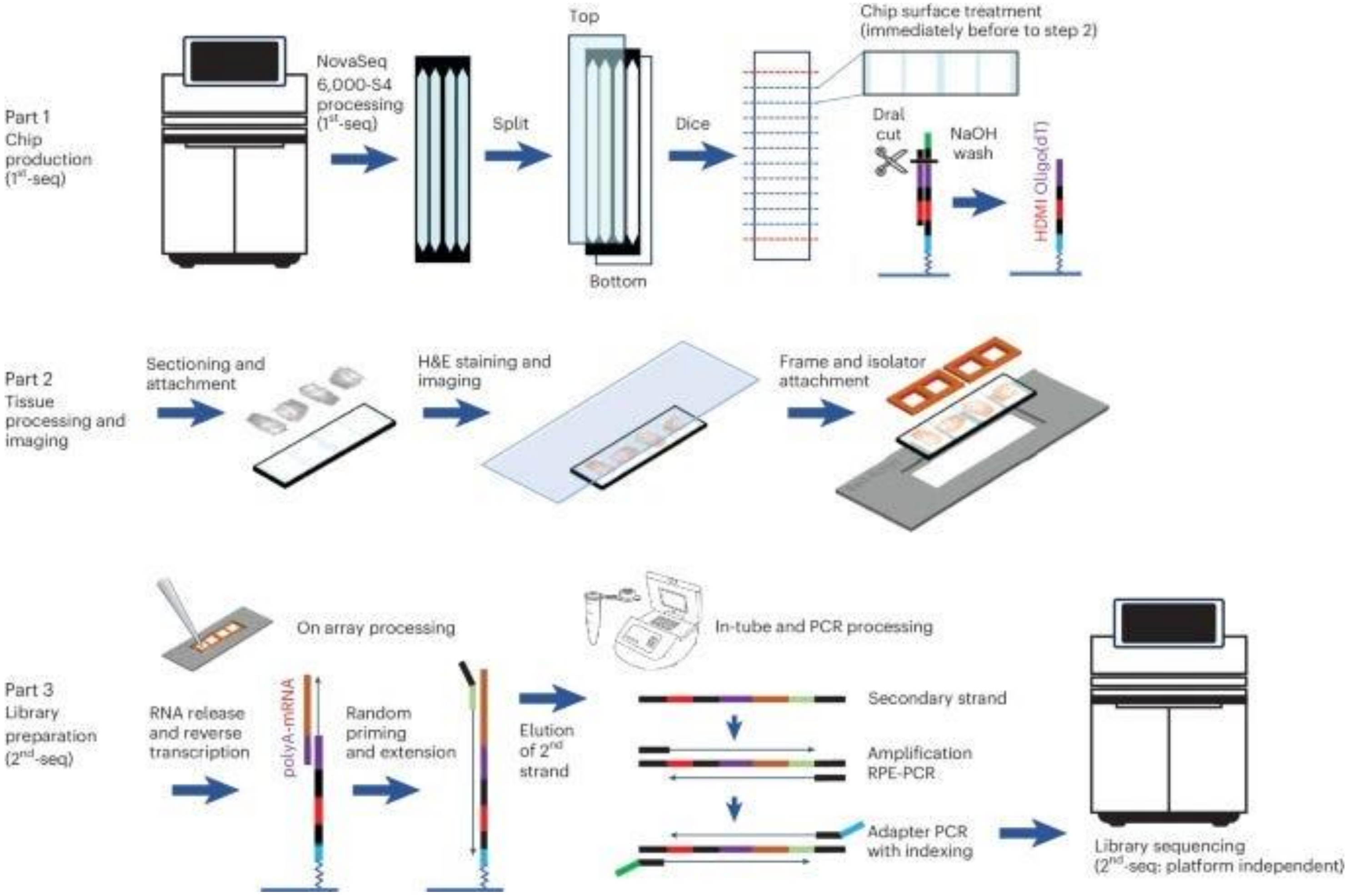


More sub-micron spatial transcriptomics on the horizon

Stereo-seq (STOmics, 220 nm)

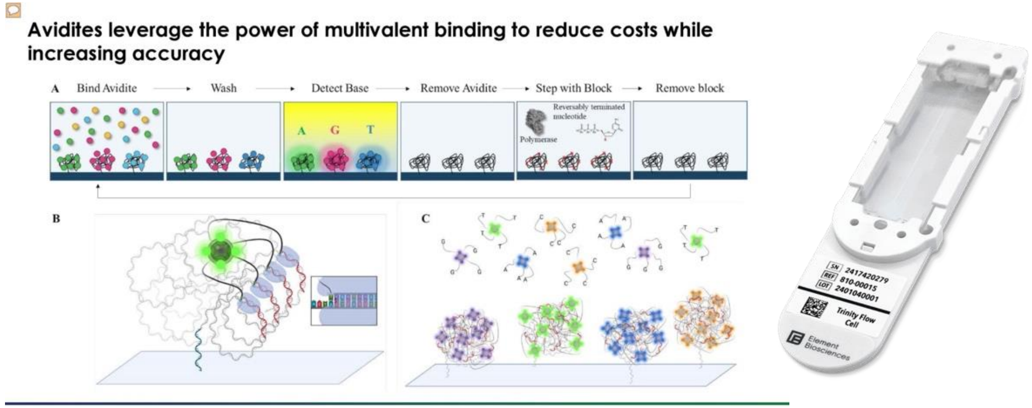


Illumina Spatial (Seq-Scope, 0.5 - 2 µm)

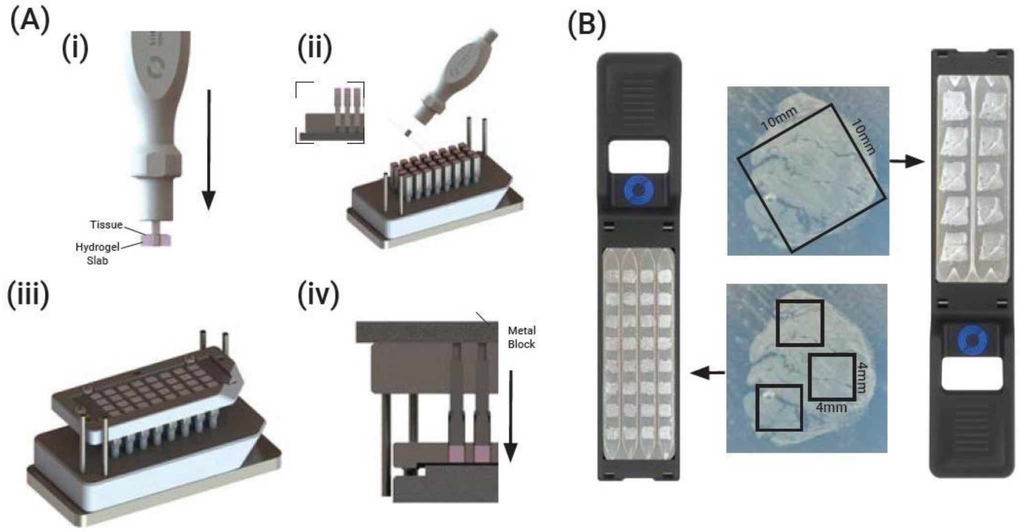


Now with better sequencing chemistries

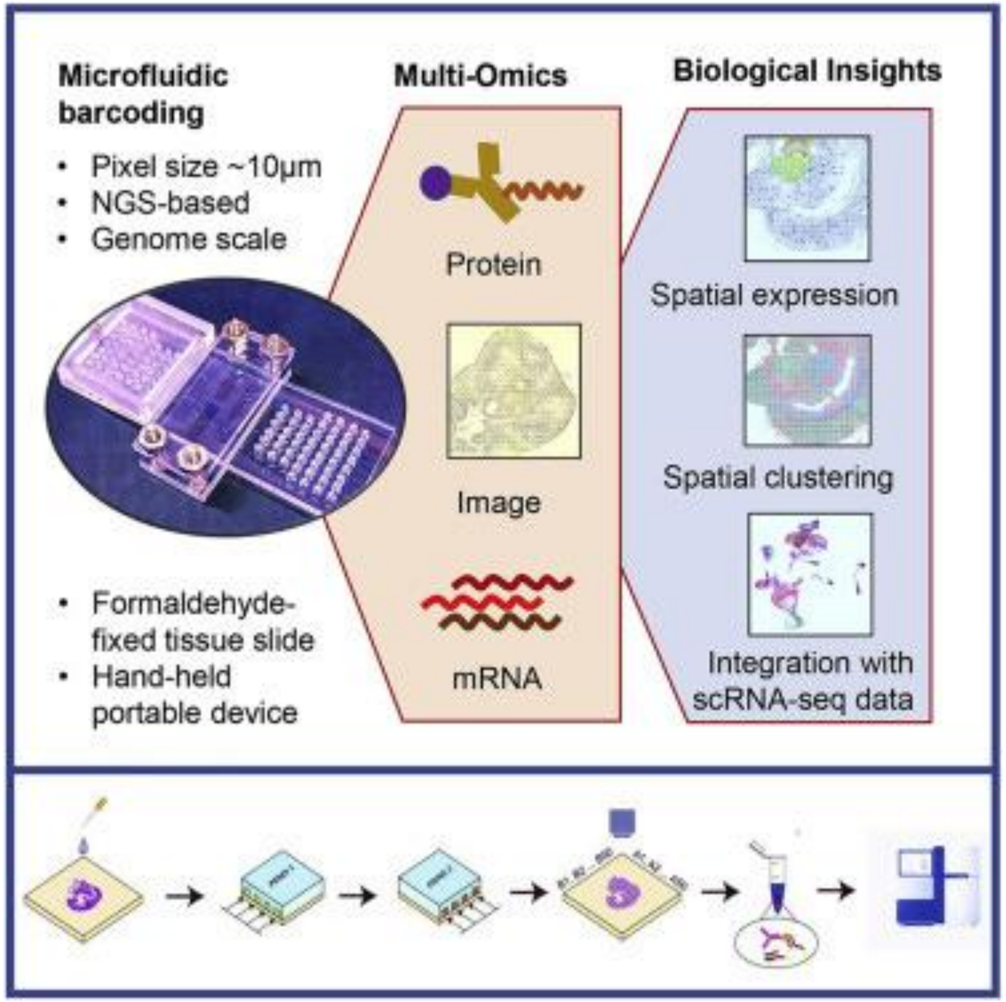
Element AVITI24

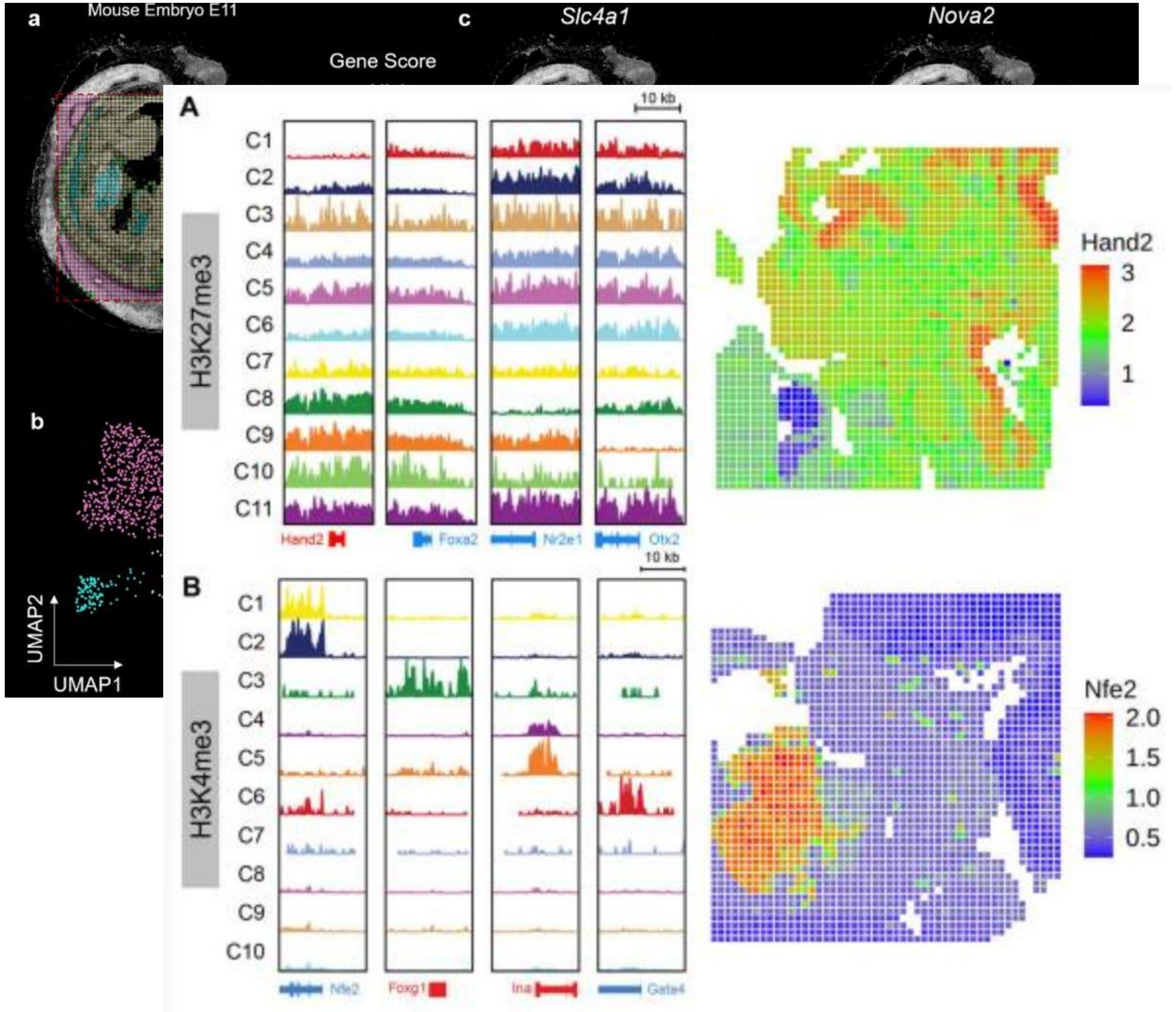


Singular Genomics G4X



Multi-omics sequencing – DBiT-seq

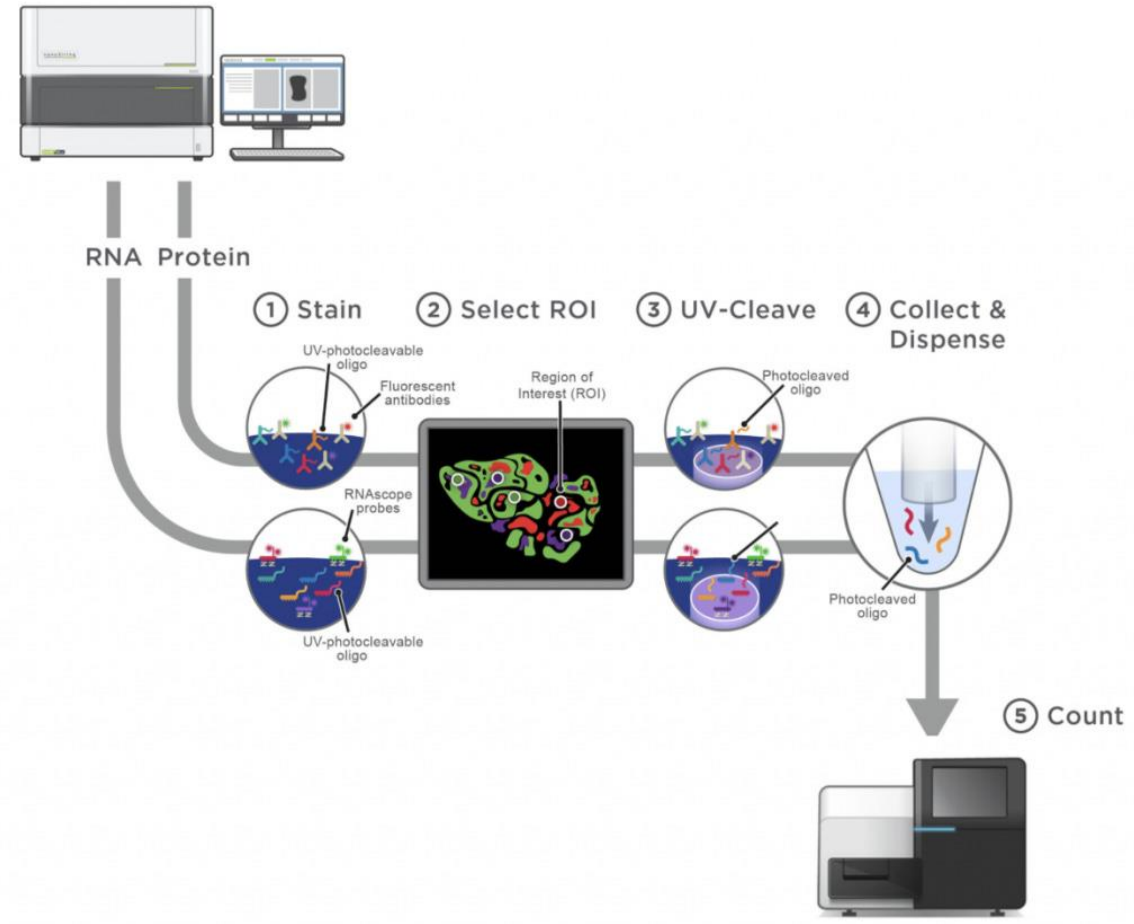


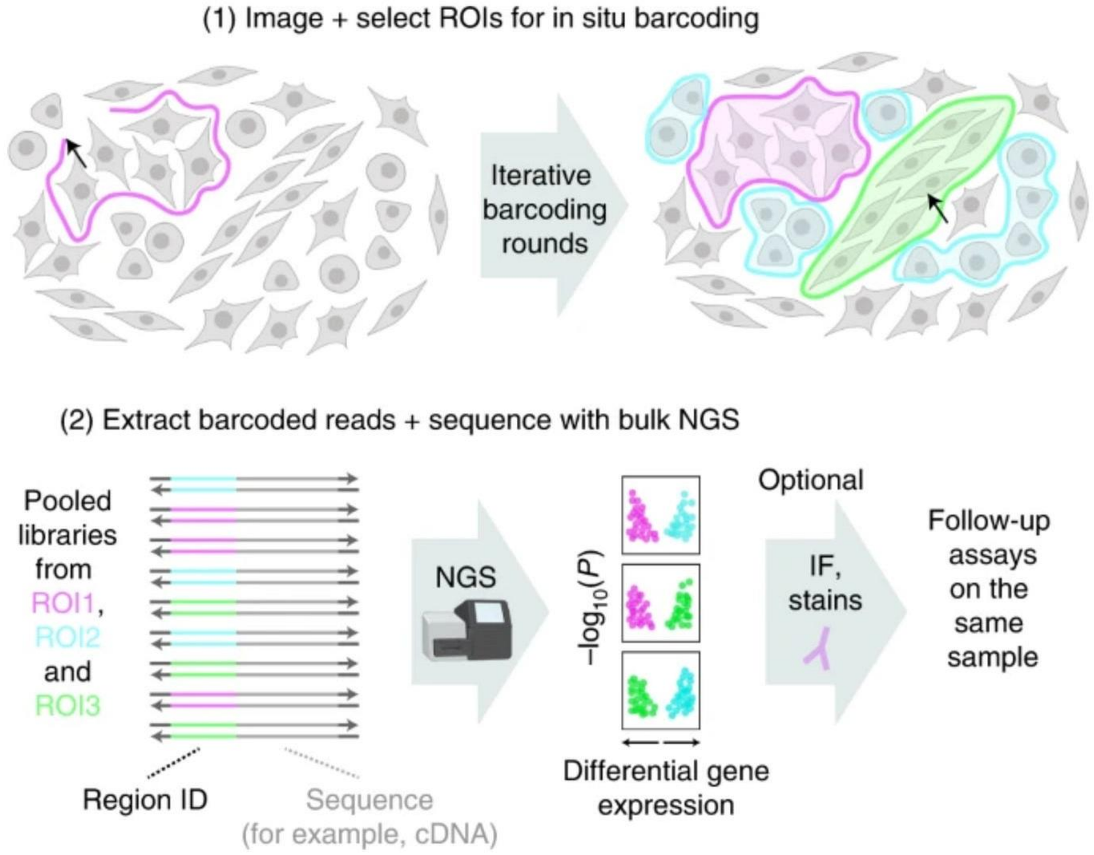


Low resolution, but only tech with epigenomics

Spatial barcoding with photo-activation

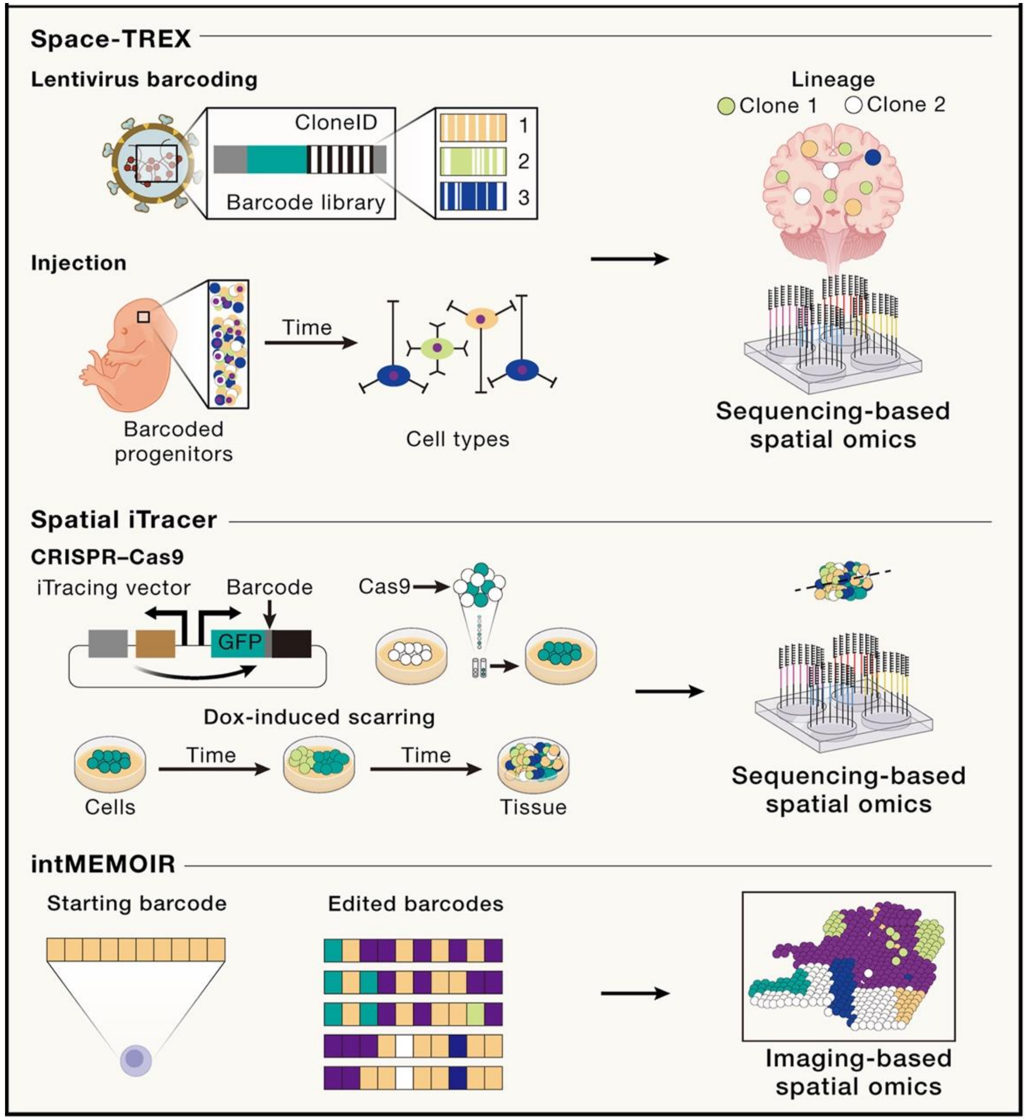
GeoMx DSP	Light-Seq



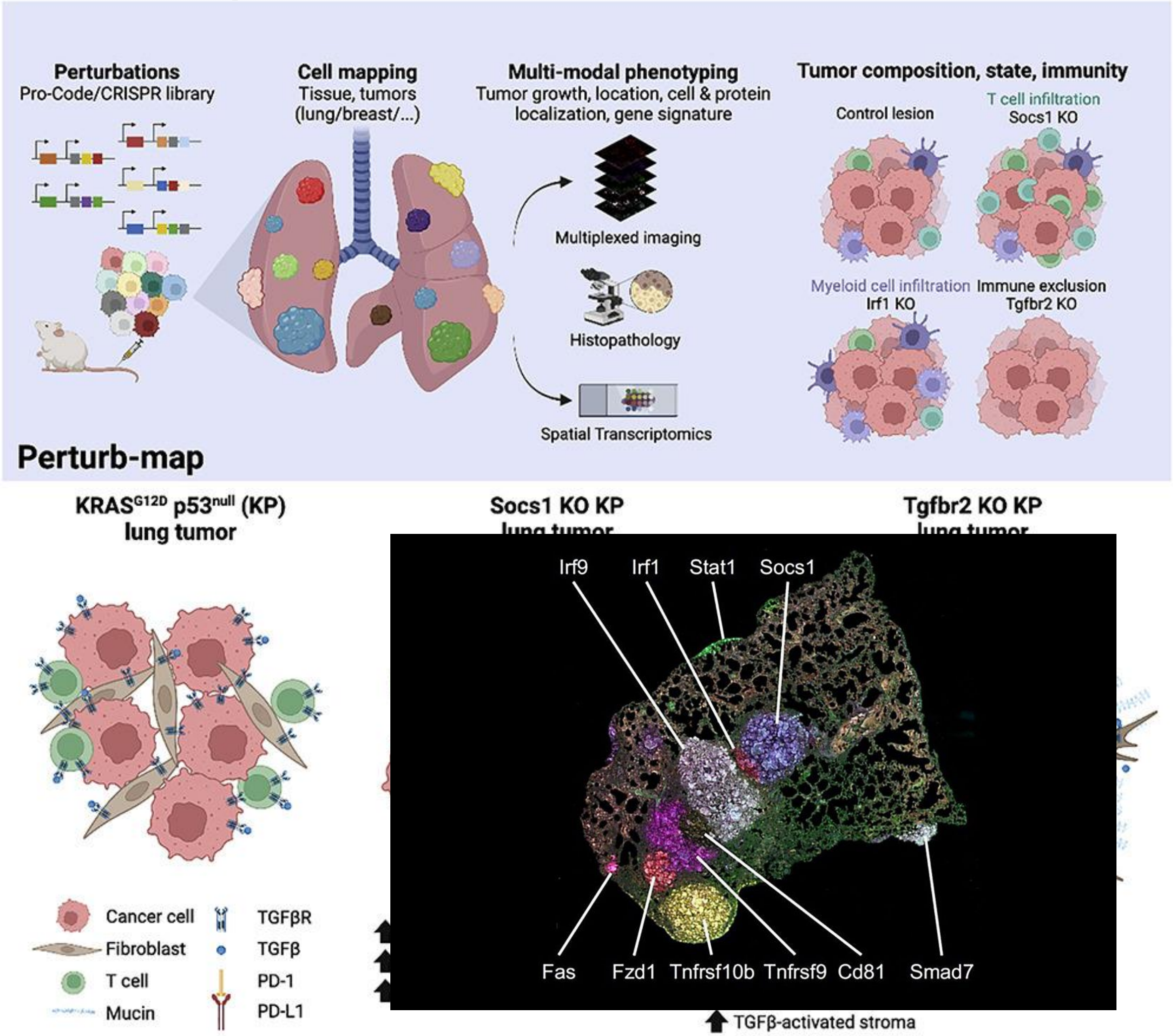


Images from NanoString GeoMx DSP; Kishi, J.Y., Liu, N., West, E.R. et al. *Nature Methods* 2022.

Temporal measurements



Perturb-Map



Liu, Longqi et al. *Cell* 2024.

Questions?

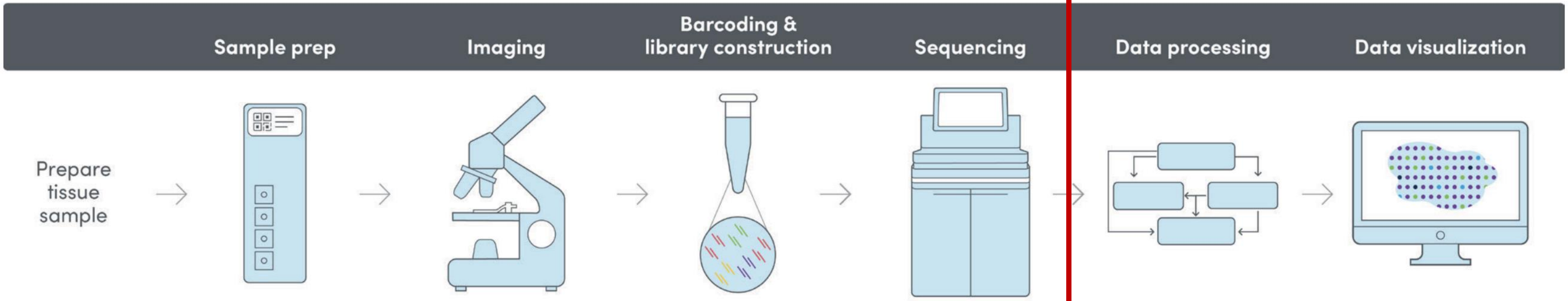
Factors to consider every step

- Why Spatial
- Cost

- Throughput
- Capture area
- Resolution

- Custom panel
- Modality
- Plex

- Controls
- Sensitivity, Specificity
- Depth

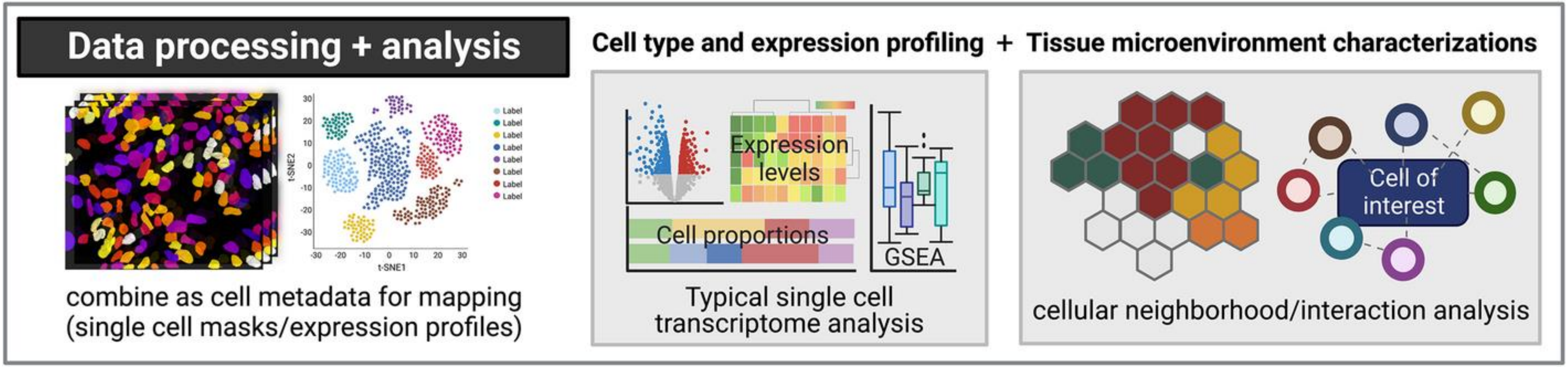


- FFPE/FF
- Size
- Tissue type
- Structure vs. RNA

- Segmentation
- Annotation
- Statistics

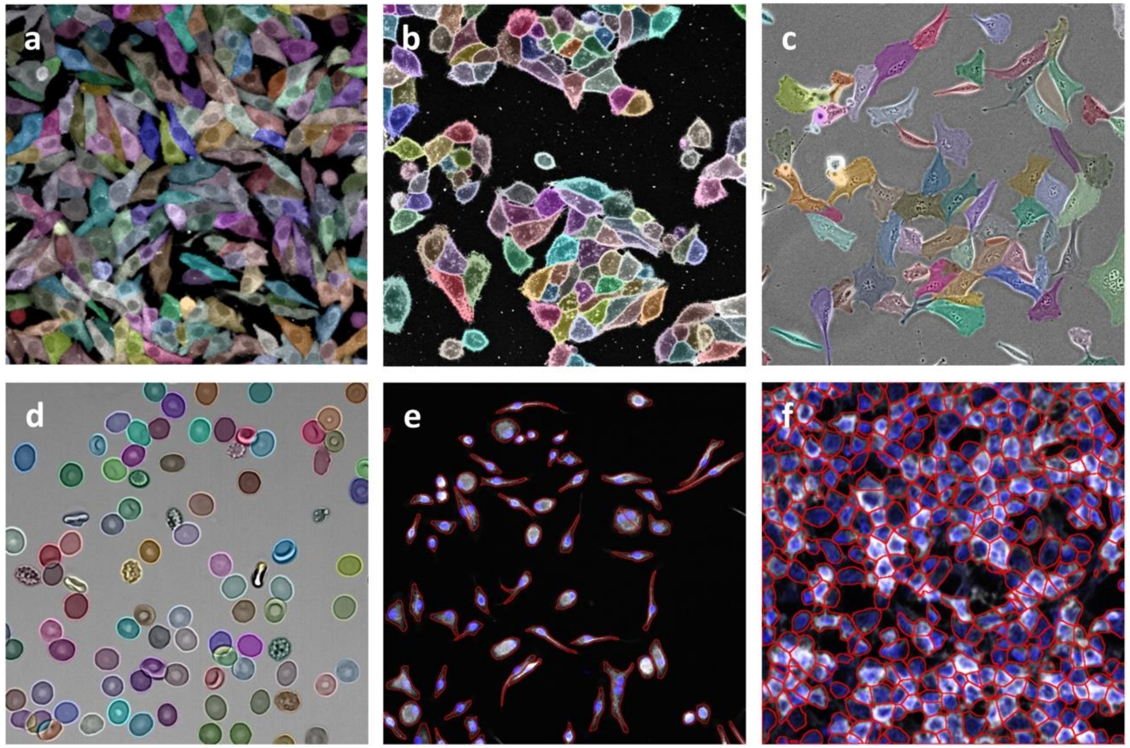
Image from 10X Genomics

Spatial omics data analysis



Park J.. *et al. Genome Biology* 2022.

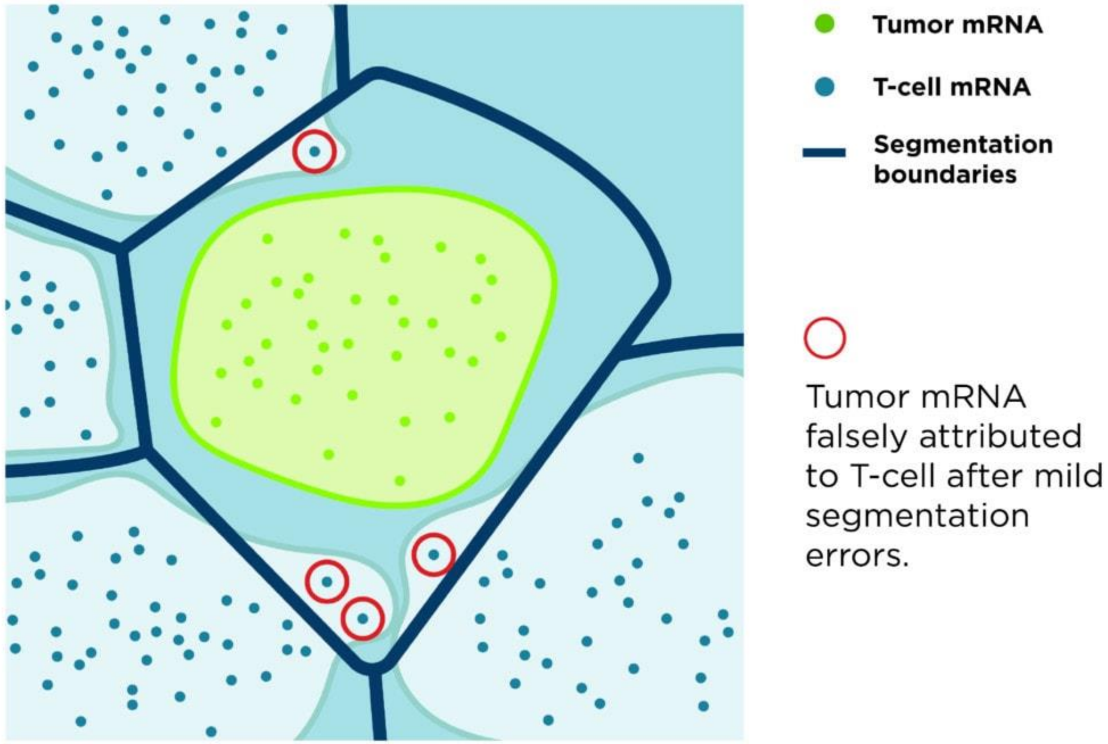
Cell segmentation – why do we care?



- **Understanding Cellular Interactions:** Cell segmentation allows scientists to explore cellular interactions in normal and pathological conditions, enhancing our understanding of fundamental biological processes.
- **Analyzing Biological Features:** Cell segmentation empowers scientists to analyze essential biological features like cell count, type, shape, and more. This analysis provides insights into how these features change over time and in response to different conditions.
- **Capturing Biologically Relevant Information:** The morphology of cells reflects their physiological state, and effective cell segmentation captures biologically relevant morphological information.
- **Accelerating Drug Discovery:** Insights gained from cell segmentation can drive drug discovery, diagnostics, and various other critical areas in biology, pharmacology, and personalized medicine.

Image from CellPose

Challenges in cell segmentation

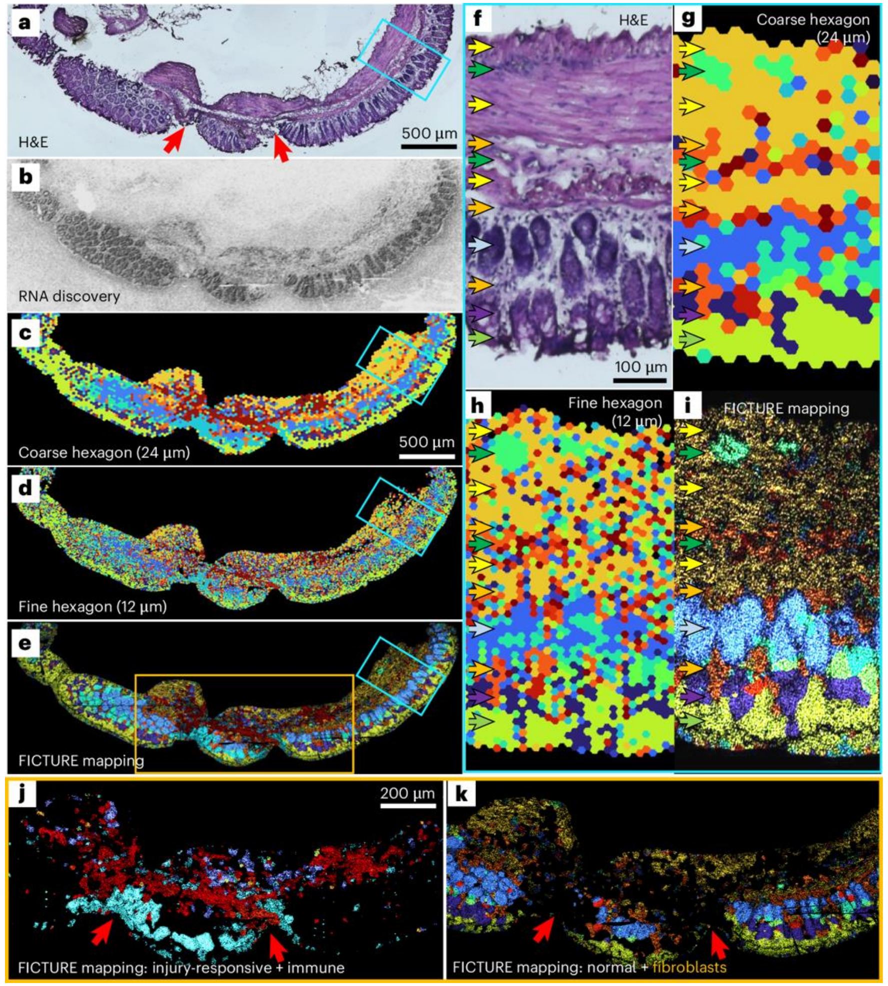


- **Heterogeneous Shapes:** Cells come in diverse and dynamic shapes, making it nearly impossible to define mathematical shape models.
- **Variation in Size and Shape:** Unlike nuclei, which are typically similar-sized structures, the cytoplasm exhibits significant variations in shape and size.
- **Weak Boundary Gradients:** Cells that are in close proximity can have weak boundary gradients, making it challenging to separate them.
- **Makeshift Nature of Segmentation Approaches:** The applicability of segmentation methods can be limited by dataset constraints, including differences in staining or imaging modality, artifacts in image capture, or morphological differences.

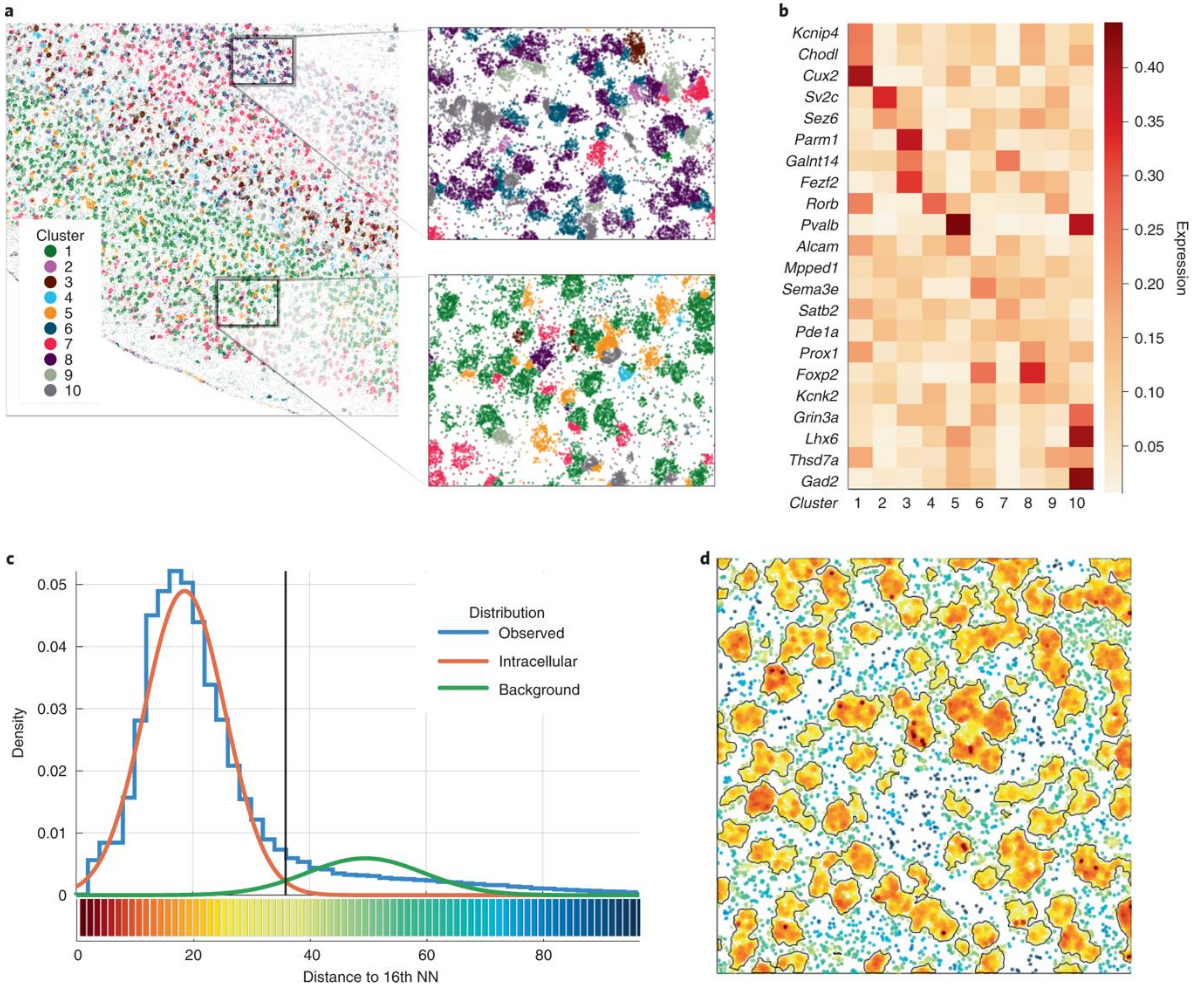
Image from Bruker NanoString Technologies

Segmentation-free methods if interaction is not important

FICTURE



Baysor



Si, Y., Lee, C., Hwang, Y. *et al.* *Nat Methods* 2024.; Petukhov, V., Xu, R.J., Soldatov, R.A. et al. *Nat Biotechnol* 2022.

Considerations for spatial data analysis workflow

	Data	Quality Control	Normalization	Dimensionality reduction	Feature exploration
scRNA-seq	formats (i.e., h5ad, Seurat .rds, .csv)	UMI (Cell QC)	Regression and/or log transformation	PCA, tSNE, UMAP	Variable genes Differential expression
Spatial	Images/segmented masks (i.e., OME-TIFFs)	FOV QC Target probe QC	(i.e., Pearson residuals for CosMx RNA data)		Spatial variability L-R/cell-cell interactions tissue architecture

QC parameters (CosMx)

Protein

Gating and comp. with neg probes

1. Neg probes

- Keep cells with neg probes between 2 and 50 (too low is indication of data preprocessing issue, too high means high background -> unreliable intensity values)

2. High expression

- Remove cells where at least X% of proteins are in the Y^{th} percentile (default X = 50, Y = 90). Likely autofluorescence cells

3. Low expression

- Remove cells where fewer than V proteins are in at least W^{th} percentile (default V = 3, W = 50). Likely not real cells.

RNA

Data modeling and filtering

1. Cell QC

- Min counts per cell (20)
- Ratio of negative to targets counts (0.1)
- Counts per unique gene (“complexity”) (1)
- Removal of unusually large or small cells (p value of Grubb’s test = 0.01)

2. Field of View (FOV) QC

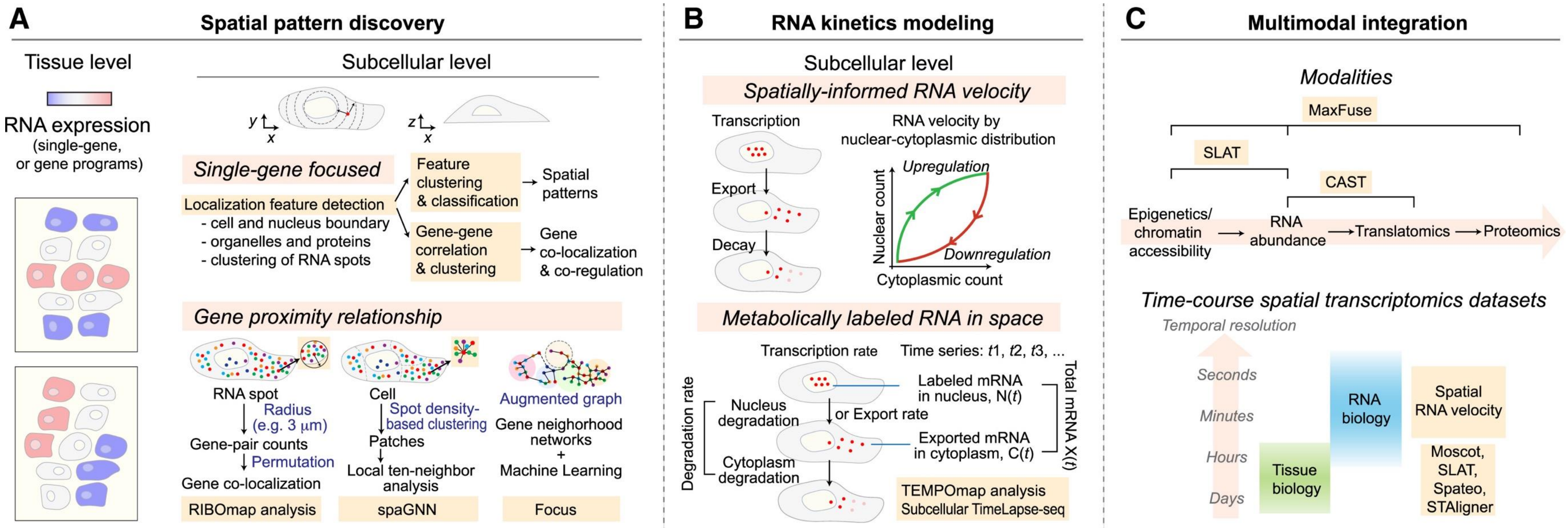
Removing low-quality FOVs: (based on mean)

- low average counts per cell (10)
- counts too close to background (0)

3. Target QC

- Score test thresholds for detecting genes above background (0.01 is the score test p-value for detection over background)
- Plot and confirm (I’m checking what AtoMx’s standard QC plot is)

Spatial analysis and data modeling



Ren J *et al. Molecular Cell* 2024.

Need for large-scale datasets in production for reference

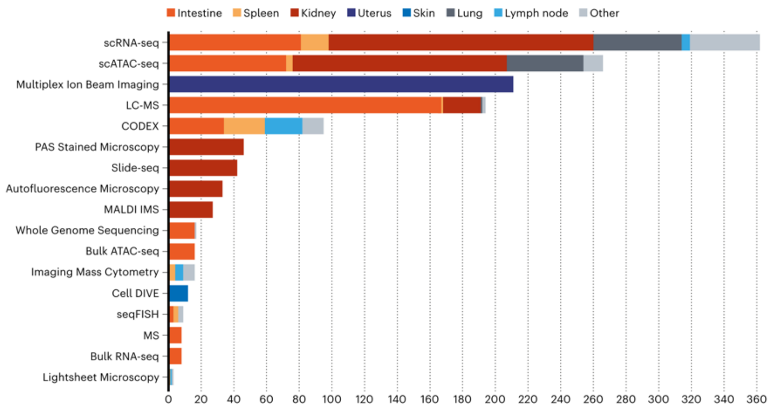
HUMAN CELL ATLAS

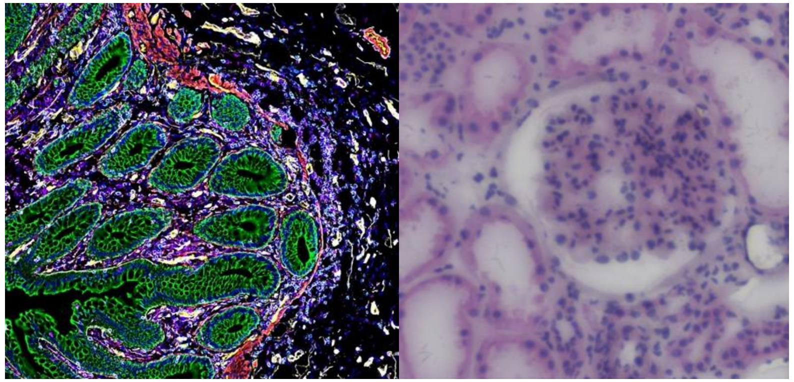
mosaic
an owkin initiative

GESTALT

STOC
SPATIOTEMPORAL OMICS CONSORTIUM

HuBMAP
The Human BioMolecular Atlas Program





HTAN
HUMAN TUMOR ATLAS NETWORK